

Realization, Extension, and Orbit Obstructions in Arithmetic Dynamics

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Abstract

We study the realization of recurrence zero sets, the extension of projective self-maps, and orbit obstructions over finite fields. We prove Derksen’s inverse realization conjecture: for every prime p , every set allowed by his classification is the zero set of a finite-order linear recurrence with constant coefficients over the fixed rational function field $\mathbb{F}_p(t)$. We also prove that a polarized projective self-map with polarization degree $d \geq 2$ over an arbitrary field extends, after a suitable closed projective embedding, to an endomorphism of projective space of the same degree. This answers the corresponding part of Poonen’s question on changing the embedding and includes nonreduced schemes. We construct two counterexamples over finite fields. For a cubic birational map of the plane, the density of points with a forward orbit defined at every step has $\liminf$ zero and $\limsup$ one over $\mathbb{F}_{2^n}$. A quintic polynomial map has a unique fixed point in every prime characteristic and admits no universal geometrically nilpotent subvariety of the kind asked for by Borisov. The two existence proofs use Frobenius reconstruction and descent along finite separable extensions, respectively. The counterexamples share a multiplicative orbit that relates the finite orders of points to the purely inseparable degrees of the projected curves over a fixed coordinate.

Keywords: arithmetic dynamics; linear recurrences; Frobenius; polarized self-maps; finite fields; geometric nilpotence.

1 Introduction

Given an algebraic map, at which times does an orbit reach a prescribed subset? Conversely, which sets of times can an orbit realize? These questions connect finitely many algebraic equations with infinitely many iterations. We study this realization problem for linear recurrences, together with two other constructions involving iteration: extending a given self-map to projective space, and making every point of a subvariety eventually reach a fixed point.

Write $\mathbb{N}_0 = \{0, 1, \dots\}$. For a self-map θ , the notation θ^n denotes its n -fold composition, with θ^0 the identity. For a rational map, successive iteration stops when the orbit first reaches the indeterminacy locus, the set of points at which the map has no regular extension. Although a variety has finitely many points over a finite field, both the number of points and the possible orbit lengths grow after extending the field. This growth leads to realization theorems as well as obstructions to density limits and uniform absorption.

1.1 From Frobenius identities to recurrence zero sets

Let p be a prime and let t be transcendental over $\mathbb{F}_p$. In characteristic p , the Frobenius map $x \mapsto x^p$ preserves addition and multiplication, so

$$(t + 1)^{p^a} = t^{p^a} + 1 \quad (a \in \mathbb{N}_0).$$

The sequence

$$u(n) = (t+1)^n - t^n - 1 \quad (n \in \mathbb{N}_0)$$

satisfies a constant-coefficient linear recurrence over $\mathbb{F}_p(t)$, and its zero set is exactly $\{p^a : a \in \mathbb{N}_0\}$. A direct verification appears in Section 2. Thus recurrence zero sets in positive characteristic can have sparse patterns governed by Frobenius powers.

Derksen classified these zero sets and asked the converse question: can every set allowed by the classification be realized by a linear recurrence over a field of characteristic p [1, Conjecture 3.6, p. 190]? The class consists of finite unions and finite modifications of finite sets, arithmetic progressions, and sets of sums of powers with rational coefficients; Definition 2.1 gives the precise conditions. Theorem 2.2 answers the question affirmatively, with the realization field fixed as $\mathbb{F}_p(t)$. The recurrence and its order may depend on the given set.

Earlier constructions cover several important cases. Corvaja–Ghioca–Scanlon–Zannier obtained a geometric realization for positive weights satisfying a size condition [2, Proposition 4.3]. Lee–Nam realized sets of sums of powers as return-time sets on split tori when every independent choice of exponents gives a nonnegative integer [3, Theorem 1.6]. Here the weights may have mixed signs, and the sum set is intersected with $\mathbb{N}_0$ after it is formed. The 2026 version of Xie–Yang includes the inverse realization problem for general recurrences in its discussion of open questions [4, §6].

The main step is to construct exponents compatible with the original weights. We place the evaluations at all elements of a finite field in a single rational function. Agreement at sufficiently many evaluation points forces a differential identity. The integer multiplicity of a specified irreducible factor is then extracted from that rational function. The differential identity controls this multiplicity modulo p , allowing Frobenius descent for the remaining coefficients. Repeating the procedure gives the desired reconstruction. Each extraction of a factor with nonzero multiplicity lowers the height of the rational function, so only finitely many such changes occur. In the projective closure, parameters may collide or tend to infinity. Ratio identities control collisions, and an integer congruence controls the parameters at infinity. A finite decomposition then removes the size restriction on the weights.

1.2 Extending the original map with its degree unchanged

Suppose that X is embedded in projective space and that $\phi : X \rightarrow X$ is a morphism. Extending its coordinate expressions to the ambient space may introduce common zeros outside X , yielding only a rational map. The extension problem asks for a choice of embedding and coordinate expressions that gives an everywhere-defined morphism of the ambient projective space.

A polarization specifies the degree to be preserved. Let k be a field and let $\mathcal{L}$ be an ample line bundle on a projective k -scheme X . Thus some positive tensor power $\mathcal{L}^{\otimes a}$ gives a closed projective embedding. Suppose that

$$\phi^* \mathcal{L} \simeq \mathcal{L}^{\otimes d}, \quad d \geq 2,$$

where the star denotes pullback and the right-hand side is a tensor power. Poonen asked whether ϕ extends to the whole projective space after changing the embedding [5, Question 1.1]. Fakhruddin established the result over infinite fields [6]. Bhatnagar–Szpiro and Poonen also proved that, for a fixed embedding, a suitable iterate extends [7, 5]. Poonen’s Remark 1.6 left open the question of changing the embedding over finite fields.

Over an arbitrary field, Theorem 3.1 constructs a closed immersion $j : X \hookrightarrow \mathbb{P}_k^M$ and a degree- d morphism $\Psi : \mathbb{P}_k^M \rightarrow \mathbb{P}_k^M$ satisfying

$$\Psi \circ j = j \circ \phi.$$

This identity gives $\Psi^n \circ j = j \circ \phi^n$ for every $n \in \mathbb{N}_0$. The embedding line bundle is a positive tensor power of $\mathcal{L}$. The construction retains nonreduced structure and allows X to be reducible or nonequidimensional.

Over a finite field, a nonempty open set of admissible coordinate tuples may have no rational point over the base field. We first choose such a tuple over a finite separable extension, then expand it in a basis over the base field. Over the algebraic closure, the expanded coordinates split into all conjugate tuples. Their common base-point-free property makes the ambient map everywhere defined. The section identities on the original scheme expand in the same basis, so the required commuting diagram survives descent. Section 3 explains the construction through squaring in a quadratic extension and treats the boundary case $d = 1$ in Proposition 3.7.

1.3 Density oscillation from a multiplicative orbit

For a rational map over a finite field, one can count the initial points that admit infinitely many successive iterations. If the base field is $\mathbb{F}_Q$, we consider their proportion among the points over $\mathbb{F}_{Q^n}$, with Q fixed and the extension degree n tending to infinity. The survey of Benedetto et al. conjectured that this proportion tends to one for every dominant rational self-map; dominance means that the image is Zariski dense in the target [8, Conjecture 18.10(b)]. In his study of the pentagram map, Weinreich questioned this assertion and conjectured that the density in that example tends to zero [9, §8].

In homogeneous coordinates $[A : B : Z]$ on the projective plane, we consider the cubic birational map

$$f[A : B : Z] = [AR : BR : A^2Z], \quad R = B(A + Z) - AZ.$$

For a finite extension $L/\mathbb{F}_Q$, let $\mathbb{P}^2(L)_f$ be the set of initial points whose forward orbit is defined at every step. Theorem 4.2 computes its cardinality and proves that

$$\delta_Q(n) = \frac{\#\mathbb{P}^2(\mathbb{F}_{Q^n})_f}{Q^{2n} + Q^n + 1}$$

fails to converge for every fixed Q . When $Q = 2$,

$$\liminf_{n \rightarrow \infty} \delta_2(n) = 0, \quad \limsup_{n \rightarrow \infty} \delta_2(n) = 1.$$

Here $\#$ denotes cardinality, and the denominator is the number of points in the projective plane. The extrema for general Q are given in (4.8).

On the open set $AZ \neq 0$, put $u = B/A$ and $v = A/Z + 1$, so that $v \neq 1$. As long as the orbit stays in this open set, its coordinates evolve by

$$S(u, v) = (u, uv), \quad S^j(u, v) = (u, u^j v) \quad (j \in \mathbb{N}_0).$$

The first coordinate is fixed, and the second is multiplied by the same element at each step. The first occurrence of $u^j v = 1$ brings the original projective orbit to an indeterminacy point. When u, v are nonzero, infinite iteration depends on whether v belongs to the finite cyclic group generated by u ; the remaining coordinate cases are counted separately. Section 4 begins with an explicit orbit over $\mathbb{F}_5$ and then expresses the point count through the sum of element orders in a cyclic group. After division by the square of the group order, this sum approaches different limits along large prime extension degrees and factorial extension degrees.

1.4 Pointwise absorption and purely inseparable degree

Finally, consider a polynomial morphism T over a finite field and a fixed point O . A subvariety is geometrically nilpotent if every geometric point eventually reaches O , with the time allowed to depend on the point; here a geometric point is a point over the algebraic closure. It is nilpotent if one common iterate sends the entire subvariety to O . Borisov constructed examples distinguishing these properties and asked whether there is a geometrically nilpotent subvariety Y such that every other geometrically nilpotent subvariety W satisfies

$$W \subseteq (T^\ell)^{-1}(T^m(Y))$$

for nonnegative integers ℓ, m that may depend on W [10, Question 13].

Theorem 5.2 gives a negative answer in every prime characteristic using the same quintic polynomial map

$$D = (y - z)y(x - z), \quad T(x, y, z) = (Dxz, Dxy, Dz^2).$$

Its unique geometric fixed point is the origin O . The surfaces

$$Y_s = \{y^{p^s} = xz^{p^s-1}\} \quad (s \geq 1)$$

are all geometrically nilpotent. For any fixed candidate Y , forward and inverse iterates cover only finitely many members of this family. The statement continues to hold when forward images are replaced by their Zariski closures.

On $xyz \neq 0$, the ratio coordinates $u = x/z$ and $v = y/z$ again give $S(u, v) = (u, uv)$. Put $K = \overline{\mathbb{F}_p}$. The nonzero geometric parameter values of Y_s belong to $K^\times$, so their multiplicative orders are finite and coprime to p . This determines absorption times point by point. The projected curve also carries the degree of its function field over $K(u)$, with the projection $(u, v) \mapsto u$ retained. Parametrize this curve by a transcendental element τ . Then $u = \tau^{p^s}$, giving

$$K(\tau^{p^s}) \subseteq K(\tau), \quad [K(\tau) : K(\tau^{p^s})] = p^s.$$

This extension is purely inseparable: each element of the larger field has a p^r -th power in the smaller field for some $r \in \mathbb{N}_0$. Section 5 proves that the degree is invariant under all integer iterates of S on the torus. It also proves that the ratio projection of a geometrically nilpotent subvariety has dimension at most one. Finitely many curve components supply only finitely many such degrees and hence cannot cover the whole family of surfaces.

Notation is introduced where it is used. Section 2 concerns recurrences and their zero sets. Section 3 uses ϕ, Ψ for the given morphism and its extension. Sections 4 and 5 use the fixed maps f, T , respectively, and share the ratio-coordinate map S above.

2 Inverse realization of zero sets of linear recurrences in positive characteristic

2.1 The problem and the classes of sets

Write $\mathbb{N}_0 = \{0, 1, \dots\}$. A constant-coefficient linear recurrence over a field K is a sequence $u : \mathbb{N}_0 \rightarrow K$ for which there are an integer $d \geq 1$ and coefficients $\alpha_0, \dots, \alpha_{d-1} \in K$, independent of n , such that

$$u(n + d) + \alpha_{d-1}u(n + d - 1) + \dots + \alpha_0u(n) = 0 \quad (n \geq 0).$$

Here d is the order of the chosen recurrence relation. Let E denote the shift operator $(Eu)(n) = u(n+1)$. The relation can then be written as $P(E)u = 0$, where $P(\lambda) = \lambda^d + \alpha_{d-1}\lambda^{d-1} + \cdots + \alpha_0$ is called a monic annihilating polynomial. Its roots are the characteristic roots; repeated roots are allowed, as is a zero characteristic root when $\alpha_0 = 0$. The zero set of the sequence is $Z(u) = \{n \in \mathbb{N}_0 : u(n) = 0\}$. We use the classes of sets introduced by Derksen [1, Definitions 1.5–1.7].

Definition 2.1. Let p be a prime, let $e, r \geq 1$ be integers, set $q = p^e$, and write $\mathbb{F}_q$ for the field with q elements. Suppose that $c_0, \dots, c_r \in \mathbb{Q}$ satisfy

$$(q-1)c_i \in \mathbb{Z} \quad (0 \leq i \leq r), \quad \sum_{i=0}^r c_i \in \mathbb{Z}, \quad c_i \neq 0 \quad (i \geq 1).$$

If at least one $c_i > 0$ for $i \geq 1$, then

$$S_q(c_0; c_1, \dots, c_r) = \left\{ c_0 + \sum_{i=1}^r c_i q^{k_i} : k_i \in \mathbb{N}_0 \right\} \cap \mathbb{N}_0$$

is called an elementary p -nested set. If a set $S \subseteq \mathbb{N}_0$ differs by finitely many elements from a finite union of elementary p -nested sets, finite sets, and infinite arithmetic progressions, then S is called p -normal. Here an infinite arithmetic progression has the form $\{a + bn : n \in \mathbb{N}_0\}$, where $a \geq 0$ and $b \geq 1$ are integers.

The exponents k_i vary independently, and the coefficients c_i may have different signs. The integrality conditions ensure that $c_0 + \sum_i c_i q^{k_i}$ is always an integer, since $c_i(q^{k_i} - 1) \in \mathbb{Z}$. Only the nonnegative values are retained.

Theorem 2.2. *For every prime p and every p -normal set $S \subseteq \mathbb{N}_0$, there is a finite-order constant-coefficient linear recurrence u over $\mathbb{F}_p(t)$ such that $Z(u) = S$. Here t is transcendental.*

Derksen [1, Conjecture 3.6] asks for a realization over some field of characteristic p ; the theorem also fixes the realization field. The recurrence order may depend on S . For positive weights, a geometric construction is given in [2, Proposition 4.3], and a finite decomposition method is given in [3]. Below we represent positive and negative weights by rational functions and examine cancellation and points at infinity in the projective closure.

A basic example. Over $\mathbb{F}_p(t)$, consider

$$u(n) = (t+1)^n - t^n - 1 \quad (n \geq 0).$$

This sequence is annihilated by $(E - (t+1))(E - t)(E - 1)$, and $Z(u) = \{p^k : k \in \mathbb{N}_0\} = S_p(0; 1)$. Indeed, $u(0) = -1 \neq 0$. For $n > 0$, write $n = p^k m$, where $p \nmid m$. Repeated use of the Frobenius identity $(x + y)^p = x^p + y^p$ in characteristic p gives

$$u(n) = ((t+1)^m - t^m - 1)^{p^k}.$$

If $m = 1$, the right-hand side is zero. If $m > 1$, the coefficient of t in the polynomial inside the parentheses is $m \neq 0$, so the right-hand side is nonzero. The zero set in this example comes from a power identity in characteristic p . General elementary sets also allow several independent powers, negative weights, and rational shifts.

The proof first realizes sums with integer weights on the integer time axis $\mathbb{Z}$, then recovers rational parameters and the restriction to nonnegative times. Intermediate expressions with negative weights are therefore defined throughout. The main step turns a fixed set of finite-field evaluations into a rational-function identity, from which the powers are reconstructed. A finite decomposition then removes the restriction relating the weights to the size of the evaluation field.

2.2 Closure properties of bi-infinite recurrences

A sequence $u : \mathbb{Z} \rightarrow K$ is called a fundamental bi-infinite recurrence if it has a monic annihilating polynomial with nonzero constant term and the recurrence relation holds for every $n \in \mathbb{Z}$. The nonzero constant term allows the recurrence to be run both forward and backward. Equivalently, it has a finite-dimensional representation

$$u(n) = \ell A^n v, \quad A \in \mathrm{GL}_a(K). \quad (2.1)$$

Here K^a is the finite-dimensional state space, $v \in K^a$ is the initial column vector, and $\ell \in (K^a)^* = \mathrm{Hom}_K(K^a, K)$ is the linear functional that reads off the sequence value; it may also be written as a row vector. The notation $\mathrm{GL}_a(K)$ denotes the group of invertible $a \times a$ matrices. A recurrence gives this representation by taking a consecutive terms as the state and using the companion matrix. Conversely, the Cayley–Hamilton theorem shows that the characteristic polynomial of A annihilates u . Write $Z_{\mathbb{Z}}(u) = \{n \in \mathbb{Z} : u(n) = 0\}$, and let $\mathcal{R}_p$ be the class of integer zero sets of fundamental bi-infinite recurrences over $\mathbb{F}_p(t)$.

Lemma 2.3. *The class $\mathcal{R}_p$ contains the empty set, $\mathbb{Z}$, and all singletons, and is closed under finite unions, integer translations, and images and preimages under affine maps with positive integer slope.*

Proof. The constant sequences 1 and 0 realize the empty set and $\mathbb{Z}$, respectively, and $t^{n-b} - 1$ realizes the singleton $\{b\}$. The termwise product of two representations (2.1) is represented by the tensor product of their state matrices, so it is again a fundamental recurrence; its zero set is the union of the two zero sets. Integer translations are obtained by changing the initial vector. An affine preimage corresponds to the subsequence $u(an + b) = \ell(A^a)^n A^b v$, whose state matrix A^a remains invertible.

Suppose that $Z_{\mathbb{Z}}(u) = B$, let $a \geq 1$ and $b \in \mathbb{Z}$, and choose $P(E)u = 0$ with $P(0) \neq 0$. Define

$$h(n) = \begin{cases} u((n-b)/a), & n \equiv b \pmod{a}, \\ 0, & n \not\equiv b \pmod{a}. \end{cases}$$

The operator E^a preserves each residue class modulo a , so $P(E^a)h = 0$. Let g be the sequence of period a that is zero on the residue class of b and one on all other residue classes. Then $(E^a - 1)g = 0$, and the zero set of $h + g$ is exactly $aB + b$. Both annihilating polynomials have nonzero constant terms, and their product annihilates $h + g$. This also covers the case $p \mid a$. $\square$

Lemma 2.4. *If u is a fundamental bi-infinite recurrence over $\mathbb{F}_{p^s}(t)$, then there is a fundamental bi-infinite recurrence over $\mathbb{F}_p(t)$ with the same zero set as u . The same conclusion holds for one-sided recurrences allowing zero characteristic roots.*

Proof. Put $L = \mathbb{F}_{p^s}(t)$ and $K = \mathbb{F}_p(t)$. The automorphisms of L/K fix t and act on the constant field; they form the Galois group $\mathrm{Gal}(L/K)$. Take the termwise norm

$$v(n) = N_{L/K}(u(n)) = \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(u(n)).$$

A product of conjugate recurrences is again a recurrence, and remains fundamental in the fundamental case. Choose a monic annihilating polynomial $H \in L[E]$ for the norm sequence v , with $H(0) \neq 0$ in the fundamental case. Since v is fixed by every σ , the polynomial $\prod_{\sigma} \sigma(H)$ is an annihilating polynomial in $K[E]$; its constant term is nonzero in the fundamental case. The field norm is zero if and only if the original element is zero. $\square$

2.3 Signed Frobenius reconstruction from finite-field evaluations

In this subsection, t remains transcendental, and ξ is a formal variable independent of t . For a nonzero rational function $R = N/D \in K(\xi)^*$, choose relatively prime polynomial numerator and denominator and define

$$H_\xi(R) = \deg_\xi N + \deg_\xi D.$$

This height counts the multiplicities of both finite zeros and finite poles after common factors have been cancelled. The following lemma reconstructs R from its evaluations at almost all $a \in \mathbb{F}_q$. The proof first differentiates with respect to t , so that the integer exponent n appears as a coefficient. The height bounds the ξ degree of the resulting polynomial, allowing sufficiently many evaluations to determine an identity.

Lemma 2.5. *Let $q = p^e > 4C$, $C \geq 1$, and $K = \mathbb{F}_q(t)$. If $R \in K(\xi)^*$, $H_\xi(R) \leq C$, and there is a set $A \subseteq \mathbb{F}_q$ with $|A| \geq q - C$ such that R is defined on A and satisfies*

$$R(a) = (t + a)^n \quad (a \in A), \quad n \in \mathbb{Z},$$

then there is a family of integers $(d_\ell)_{\ell \geq 0}$, with only finitely many nonzero d_ℓ , such that

$$R(\xi) = \prod_{\ell \geq 0} (\xi + t^{q^\ell})^{d_\ell}, \quad n = \sum_{\ell \geq 0} d_\ell q^\ell. \quad (2.2)$$

Proof. At each step, taking p th roots of the coefficients also requires a Frobenius permutation of the evaluation points. Let k_j be the accumulated number of permutations at step j , reduced modulo e ; we call it the Frobenius phase of that step. Starting from $(R_0, n_0, A_0) = (R, n, A)$, we maintain

$$R_j(a) = (t + a^{p^{k_j}})^{n_j} \quad (a \in A_j), \quad k_j = j \bmod e, \quad 0 \leq k_j < e. \quad (2.3)$$

Let $C_j = H_\xi(R_j) \leq C$ and write $R_j = N_j/D_j$ in relatively prime form. Let ∂_t denote the formal derivative fixing ξ and $\mathbb{F}_q$. Differentiating (2.3) with respect to t and clearing denominators shows that the polynomial

$$(t + \xi^{p^{k_j}})(\partial_t N_j D_j - N_j \partial_t D_j) - n_j N_j D_j$$

vanishes on A_j . Its ξ degree is at most $C_j + p^{k_j} \leq C + q/p < q - C \leq |A_j|$, so it is identically zero. Thus

$$\partial_t \log R_j = \frac{n_j}{t + \xi^{p^{k_j}}}. \quad (2.4)$$

Here $\partial_t \log R_j$ denotes the logarithmic derivative $(\partial_t R_j)/R_j$. The polynomial $f_j = t + \xi^{p^{k_j}}$ is irreducible in $\mathbb{F}_q(t)[\xi]$: apply Eisenstein's criterion at the prime element t in $\mathbb{F}_q[t][\xi]$. Let $e_j = \text{ord}_{f_j}(R_j)$ be the integer multiplicity of f_j in the numerator minus its multiplicity in the denominator, and write $R_j = f_j^{e_j} S_j$, where neither the numerator nor the denominator of S_j is divisible by f_j . The logarithmic derivative $\partial_t \log S_j$ is regular at f_j , and $\partial_t f_j = 1$. Equation (2.4) gives

$$e_j \equiv n_j \pmod{p}, \quad \partial_t S_j = 0, \quad S_j \in \mathbb{F}_q(t^p)(\xi). \quad (2.5)$$

The last assertion about the kernel of the derivation follows by viewing t as transcendental over $\mathbb{F}_q(\xi)$. The derivative determines only the residue of e_j modulo p ; the e_j used in the recurrence is always the signed integer multiplicity. Removing the entire multiplicity of f_j gives

$$H_\xi(S_j) = C_j - |e_j|p^{k_j}.$$

Let $\Phi : \mathbb{F}_q(t)(\xi) \longrightarrow \mathbb{F}_q(t^p)(\xi)$ be the isomorphism that raises coefficients to the p th power and fixes the variable ξ . Define

$$R_{j+1} = \Phi^{-1}(S_j), \quad n_{j+1} = (n_j - e_j)/p, \quad A_{j+1} = \{a \in \mathbb{F}_q : a^p \in A_j\}.$$

Taking roots of coefficients preserves relative primeness and ξ degrees. For $a \in A_{j+1}$,

$$R_{j+1}(a)^p = S_j(a^p) = (t + a^{p^{k_j+1}})^{n_j - e_j}.$$

Frobenius is injective on a field, so this gives (2.3) at the next step. When $k_j = e - 1$, the identity $a^{p^e} = a$ returns the phase to zero. We also have

$$|A_{j+1}| = |A_j|, \quad C_{j+1} = C_j - |e_j|p^{k_j}. \quad (2.6)$$

Since the C_j are nonnegative integers, only finitely many e_j can be nonzero. Choose J beyond the last nonzero term. Then $n_{J+h} = n_J/p^h \in \mathbb{Z}$ for every $h \geq 0$, so $n_J = 0$. Now $R_J(a) = 1$ at more than C_J points, and clearing denominators gives $R_J = 1$.

Working backward from $R_j = f_j^{e_j} \Phi(R_{j+1})$, we obtain

$$\begin{aligned} R_0 &= \prod_{j < J} (t^{p^j} + \xi^{p^{k_j}})^{e_j} \\ &= \prod_{j < J} (t^{q^{\lfloor j/e \rfloor}} + \xi)^{e_j p^{k_j}}. \end{aligned}$$

Set

$$d_\ell = \sum_{\substack{0 \leq k < e \\ e\ell + k < J}} p^k e_{e\ell + k}.$$

Combining terms with the same $\ell = \lfloor j/e \rfloor$ gives the first identity in (2.2). Since all e_j, n_j are integers, the relations $n_j = e_j + pn_{j+1}$ and $n_J = 0$ give the integer identity

$$n = n_0 = \sum_{j < J} p^j e_j = \sum_{\ell \geq 0} d_\ell q^\ell,$$

which is the second identity. □

The entire descent takes place in rational function fields. For example, when $k_j > 0$, the factor $t + \xi^{p^{k_j}}$ has a repeated root over an algebraic closure, and the proof still uses its integer valuation in $\mathbb{F}_q(t)[\xi]$. The phase returns to zero every e steps, combining the successive p th roots into the q th powers in the conclusion. The resulting levels ℓ are nonnegative, while the multiplicities d_ℓ may have either sign.

For example, if $q > 8$, the rational function of height two

$$R(\xi) = \frac{\xi + t^{q^2}}{\xi + t^q}$$

satisfies $R(a) = (t + a)^{q^2 - q}$ at every $a \in \mathbb{F}_q$. It corresponds to $d_2 = 1$ and $d_1 = -1$; the factor in the denominator records the negative weight. These evaluation identities follow directly from $a^q = a$.

2.4 Geometric realization of signed weights

We now apply the reconstruction lemma to the construction of zero sets. Keeping one coordinate for each $a \in \mathbb{F}_q$ records the evaluations over the whole finite field simultaneously. In the projective closure, parameters may coincide or tend to infinity. The proof below treats cancellation among finite parameters and the total weight at infinity separately, assigning a nonnegative exponent to each original weight from the geometric conditions.

Proposition 2.6. *Let $r \geq 1$, let $c_1, \dots, c_r$ be nonzero integers, set $C = \sum_i |c_i|$, and suppose that $q = p^e > 4C$. Then*

$$E_q(\mathbf{c}) = \left\{ \sum_{i=1}^r c_i q^{k_i} : k_i \geq 0 \right\} \subseteq \mathbb{Z}$$

belongs to $\mathcal{R}_p$.

Proof. We first work over $K = \mathbb{F}_q(t)$. Let $\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$ be the multiplicative algebraic group; for any extension field L , its L -valued points form L^* , with multiplication as the group law. Take the coordinate torus $\mathcal{T} = (\mathbb{G}_m)^{\mathbb{F}_q}$, defined over $\mathbb{F}_q$, and the point $P = (t + a)_{a \in \mathbb{F}_q} \in \mathcal{T}(K)$. The superscript $\mathbb{F}_q$ means that the coordinates are indexed by the q elements of this finite field, so $\mathcal{T}(L) = (L^*)^q$; the integer power P^n is taken coordinatewise. On the open subset $U = (\mathbb{A}^1 \setminus (-\mathbb{F}_q))^r$ of $Y = (\mathbb{P}^1)^r$, define

$$\nu(\mathbf{y})_a = \prod_i (y_i + a)^{c_i}.$$

Let $\Gamma \subseteq Y \times \mathcal{T}$ be the reduced closure of the graph of ν , and let $V \subseteq \mathcal{T}$ be its reduced closed image under projection. Since Y is projective, this projection is proper: its image remains closed after every field extension and equals the underlying space of V . All objects are defined over $\mathbb{F}_q$. Choose $\zeta \in \mathbb{F}_q^*$ of order $q - 1$. We prove that, for every integer n ,

$$n \in E_q(\mathbf{c}) \iff P^n \in V \text{ and } \zeta^n = \zeta^{\sum_i c_i}. \quad (2.7)$$

For the forward implication, set $y_i = t^{q^{k_i}}$. Since $a^{q^{k_i}} = a$, we have $y_i + a = (t + a)^{q^{k_i}}$, so $\nu(\mathbf{y}) = P^n$. Moreover, $q^{k_i} \equiv 1 \pmod{q - 1}$, so the additional ζ coordinate also satisfies the required condition.

Conversely, suppose that the right-hand side of (2.7) holds. Fix an algebraic closure $\overline{K}$ of K and choose an $\overline{K}$ -valued point of Γ above P^n . Write its Y coordinate as $\mathbf{y} = (y_1, \dots, y_r)$. Set

$$A = \{a \in \mathbb{F}_q : \text{every finite } y_i \text{ is different from } -a\}.$$

Then $|A| \geq q - r \geq q - C$. Fix $b \in A$. For each $a \in A$, the function

$$\prod_i \left(\frac{y_i + a}{y_i + b} \right)^{c_i}$$

is regular and invertible on a neighborhood of $\mathbf{y}$. Indeed, near $y_i = \infty$, use the local coordinate $z_i = 1/y_i$. The corresponding ratio is $(1 + az_i)/(1 + bz_i)$, which takes the value one at $z_i = 0$. The ratio identity on the graph therefore extends to this point of Γ . Since its torus coordinate is P^n , the displayed product has value $(t + a)^n / (t + b)^n$. Set

$$F(\xi) = \prod_{i: y_i \neq \infty} (\xi + y_i)^{c_i}, \quad R(\xi) = \frac{(t + b)^n}{F(b)} F(\xi). \quad (2.8)$$

Then $R \in \overline{K}(\xi)^*$, $H_\xi(R) \leq C$, and $R(a) = (t + a)^n$ for every $a \in A$.

The reconstruction lemma requires the coefficients to lie in K ; this follows from the same evaluation data. Write $R = N/D$ in relatively prime form, and let $a_0 = \deg N$ and $b_0 = \deg D$. For unknown polynomials N_0, D_0 , consider the homogeneous linear system over K

$$N_0(a) - (t+a)^n D_0(a) = 0 \quad (a \in A), \quad \deg N_0 \leq a_0, \quad \deg D_0 \leq b_0.$$

This system has a nonzero solution over $\overline{K}$. The rank of a linear system is unchanged by field extension, so it also has a nonzero solution over K . We have $D_0 \neq 0$, since otherwise N_0 would have more roots than its degree and would also be zero. The polynomial $ND_0 - N_0D$ has degree at most $a_0 + b_0 \leq C < |A|$ and vanishes on A , so $R = N_0/D_0 \in K(\xi)$. This linear argument also applies to purely inseparable extensions.

By Lemma 2.5,

$$R(\xi) = \prod_{\ell} (\xi + t^{q^\ell})^{d_\ell}, \quad n = \sum_{\ell} d_\ell q^\ell.$$

In this product, d_ℓ records the combined weight; we must still assign exponents to the individual original weights c_i . Group the finite y_i by equality of their values. If a group I has nonzero net weight $s_I = \sum_{i \in I} c_i$, comparison of the factors in (2.8) shows that its common parameter must be some t^{q^ℓ} ; assign exponent ℓ to every original label in this group. Assign exponent zero to all labels in a group of net weight zero, since its contribution is $\sum_{i \in I} c_i q^0 = s_I = 0$. For example, weights 2, -1, -1 at the same parameter value give a product of factors equal to one. Assigning the same exponent to all three original weights keeps their contribution zero.

Parameters at infinity contribute one to the ratios and hence have not appeared among the factors of F . Let $I_\infty = \{i : y_i = \infty\}$. Comparing the difference between the numerator and denominator degrees gives

$$n \equiv \sum_{\ell} d_\ell = \sum_{i \notin I_\infty} c_i \pmod{q-1}.$$

The congruence condition in (2.7) now gives $\sum_{i \in I_\infty} c_i \equiv 0 \pmod{q-1}$. The absolute value of this integer is at most $C < q-1$, so it is zero. Assigning exponent zero to these labels as well yields $n = \sum_i c_i q^{k_i}$ with all $k_i \geq 0$. This proves (2.7) while retaining every original weight. The ζ coordinate determines the total weight at infinity: the congruence modulo $q-1$, together with $C < q-1$, forces that integer to be zero.

Finally, we convert the fixed geometric conditions into a recurrence. The coordinate ring of the torus consists of Laurent polynomials, which allow negative integer powers of the variables. This ring is Noetherian, so finitely many such equations define V . Choose a finite set of Laurent polynomial equations for V and add the equation $z - \zeta^{\sum_i c_i} = 0$ in a further variable z ; denote the equations by $f_1, \dots, f_s$. Each sequence

$$a_j(n) = f_j(P^n, \zeta^n)$$

is a finite linear combination over $\mathbb{F}_q(t)$ of exponential terms with nonzero bases. More explicitly, for integers m_a, h , the coordinate monomial $\prod_a x_a^{m_a} z^h$ takes the value $(\prod_a (t+a)^{m_a} \zeta^h)^n$ at (P^n, ζ^n) , and its base is nonzero. Thus each a_j is a fundamental bi-infinite recurrence. To obtain a single sequence whose zero set is the common zero set of these equations, choose a basis $\beta_1, \dots, \beta_s$ of $\mathbb{F}_{q^s}/\mathbb{F}_q$. These elements remain linearly independent over $\mathbb{F}_q(t)$. Hence the zero set of $a(n) = \sum_j \beta_j a_j(n)$ is exactly the common zero set of all the a_j . Descending a to $\mathbb{F}_p(t)$ by Lemma 2.4 proves the claim. $\square$

2.5 Arbitrary weights and recovery of the original parameters

Proposition 2.6 requires the finite field to have more than four times the sum of the absolute values of the weights. For a fixed base q_0 , grouping the exponents modulo m replaces the base

by $Q = q_0^m$, but the residues also enter the weights. We use the circular-gap decomposition of Lee–Nam [3] to choose a common starting point for the residues, making the new weights small enough relative to Q . The lower bounds on the exponents arising from the common shift are handled by induction on the number of terms.

Proposition 2.7. *For every $q_0 = p^f$ with $f \geq 1$ and all nonzero integers $c_1, \dots, c_r$, the set $E_{q_0}(\mathbf{c})$ belongs to $\mathcal{R}_p$.*

Proof. We induct on the number of terms r , with the convention that the value set of the empty sum is $\{0\}$. The case $r = 0$ follows from Lemma 2.3. Suppose that $r \geq 1$ and set $C = \sum_i |c_i|$. Choose a positive integer m such that

$$4Cq_0^{-\lceil m/r \rceil} < 1, \quad Q = q_0^m.$$

Write the exponents as $k_i = m\ell_i + \rho_i$, where $0 \leq \rho_i < m$ and $\ell_i \geq 0$. There are only finitely many residue vectors $(\rho_1, \dots, \rho_r)$. For a fixed residue vector, regard the ρ_i as integer positions on a circle of circumference m . Among at most r distinct points, there is a gap of length at least $\lceil m/r \rceil$; when there is only one point, take the whole circle as the gap. Let a be the first point after this gap. Starting the residues at a places all points in an interval of length at most $m - \lceil m/r \rceil$. Explicitly, set

$$w_i = (\rho_i - a) \bmod m, \quad \varepsilon_i = \mathbf{1}_{\rho_i < a}.$$

Here ε_i is one if $\rho_i < a$ and zero otherwise. Thus $0 \leq w_i \leq m - \lceil m/r \rceil$ and $\rho_i = a + w_i - m\varepsilon_i$. First consider all tuples satisfying $\ell_i \geq \varepsilon_i$. Substituting $\rho_i = a + w_i - m\varepsilon_i$ gives $q_0^{\rho_i} Q^{\ell_i} = q_0^a q_0^{w_i} Q^{\ell_i - \varepsilon_i}$, and the $\ell_i - \varepsilon_i$ range independently over the nonnegative integers. The value set of this part is therefore

$$\left\{ \sum_i c_i q_0^{\rho_i} Q^{\ell_i} : \ell_i \geq \varepsilon_i \right\} = q_0^a E_Q(c_1 q_0^{w_1}, \dots, c_r q_0^{w_r}). \quad (2.9)$$

The sum of the absolute values of the new weights is at most $Cq_0^{m - \lceil m/r \rceil} < Q/4$, so Proposition 2.6 and closure under affine images realize (2.9).

Every remaining tuple has an index with $\varepsilon_i = 1$ and $\ell_i = 0$. Fix one such i and let all other exponents vary independently. The resulting value set is

$$c_i q_0^{\rho_i} + E_Q((c_j q_0^{\rho_j})_{j \neq i}).$$

It is realized by the induction hypothesis for $r - 1$ terms and an integer translation. Taking the union over all i with $\varepsilon_i = 1$ covers exactly the remaining tuples. Finally, take the union over the finitely many residue vectors. Each time an exponent is fixed, the number of terms decreases by one, so the induction requires only finitely many steps. $\square$

Proof of Theorem 2.2. We first recover rational weights and the constant term by taking an affine preimage with positive integer slope. For an elementary set in Definition 2.1, put $D = q - 1$ and $d_i = Dc_i \in \mathbb{Z}$. Proposition 2.7 gives a fundamental bi-infinite recurrence a over $\mathbb{F}_p(t)$ such that $\mathbb{Z}\mathbb{Z}(a) = E_q(d_1, \dots, d_r)$. Then $b(n) = a(Dn - d_0)$ is again a fundamental recurrence, and for every $n \in \mathbb{Z}$,

$$b(n) = 0 \iff n \in \left\{ c_0 + \sum_i c_i q^{k_i} : k_i \geq 0 \right\}.$$

Restricting to $\mathbb{N}_0$ realizes the desired elementary set.

For an infinite arithmetic progression, first realize its residue class by a periodic sequence, then modify a finite initial segment. Finite sets are realized as finite unions of singleton zero

sets, and finite unions are realized by termwise products. Finally, to make a finite symmetric-difference modification, set the sequence value to zero at each zero to be added and to one at each zero to be removed. The difference between the modified and original sequences has finite support and is annihilated by some E^N . If the original sequence is annihilated by $P(E)$, the modified sequence is annihilated by $E^N P(E)$. Thus all finite modifications still give one-sided linear recurrences. All operations take place in the fixed field $\mathbb{F}_p(t)$. $\square$

Corollary 2.8. *The zero sets of finite-order constant-coefficient linear recurrences over $\mathbb{F}_p(t)$ are exactly the p -normal sets.*

Proof. One direction is Theorem 2.2; the other is Derksen's forward theorem [1, Theorem 1.8]. $\square$

Corollary 2.9. *Every p -normal set is exactly the set of nonnegative times at which an orbit of some finite-dimensional linear self-map over $\mathbb{F}_p(t)$ lies in a fixed hyperplane.*

Proof. Choose an annihilating relation of order d for the recurrence in Theorem 2.2, in the form given at the beginning of this section. On K^d , with $K = \mathbb{F}_p(t)$, define

$$A(x_0, \dots, x_{d-1}) = (x_1, \dots, x_{d-1}, -\sum_{i=0}^{d-1} \alpha_i x_i).$$

Set $v = (u(0), \dots, u(d-1))$. Then $A^n v = (u(n), \dots, u(n+d-1))$. For the hyperplane $H = \{(x_0, \dots, x_{d-1}) : x_0 = 0\}$, we have $A^n v \in H$ if and only if $u(n) = 0$. Allowing zero characteristic roots permits A to be noninvertible and therefore retains finite modifications of the initial segment. $\square$

3 Extending polarized self-maps with the same degree

Suppose a projective scheme is embedded in projective space. A self-map can be described by sections on the scheme, but lifting those sections to homogeneous polynomials on the ambient space may introduce common zeros. The extension problem asks for an embedding and lifts that remove all such common zeros while preserving the polarization degree.

Let k be a field, let X be a projective k -scheme, let $\mathcal{L}$ be an ample line bundle on X , and let $\phi : X \rightarrow X$ be a k -morphism satisfying

$$\phi^* \mathcal{L} \simeq \mathcal{L}^{\otimes d}. \quad (3.1)$$

Here ϕ^* denotes pullback of line bundles, and $\mathcal{L}^{\otimes d}$ is the d -fold tensor power. The integer d is positive, and (3.1) is called a polarization of degree d . For a projective scheme, ampleness ensures that some positive tensor power is very ample, meaning that its global sections give a closed projective embedding. Write $H^0(X, \mathcal{L})$ for the space of global sections, $\mathcal{O}_{\mathbb{P}^N}(1)$ for the standard hyperplane line bundle, and $\mathcal{O}_{\mathbb{P}^N}(d) = \mathcal{O}_{\mathbb{P}^N}(1)^{\otimes d}$. The global sections of the latter are homogeneous polynomials of degree d . A tuple of forms of the same degree has no geometric base point if its members have no common projective zero over the algebraic closure $\bar{k}$. Such a tuple defines an everywhere-defined morphism of projective space.

We construct a closed immersion $j : X \hookrightarrow \mathbb{P}_k^M$ and an ambient morphism $\Psi : \mathbb{P}_k^M \rightarrow \mathbb{P}_k^M$ such that

$$\Psi \circ j = j \circ \phi. \quad (3.2)$$

This equality gives $\Psi^m j = j \phi^m$ for the same j and Ψ and every $m \geq 0$.

For $d \geq 2$, Fakhruddin proved the extension theorem over infinite fields, allowing a change of embedding [6, Proposition 2.1 and Corollary 2.3]. Bhatnagar-Szpiro proved that some iterate

extends for a fixed embedding [7, Theorem 2.1]. Poonen extended this result on iterates to arbitrary fields [5, Theorem 1.4] and posed the question of changing the embedding over finite fields [5, Question 1.1 and Remark 1.6]. The following theorem extends the original map and preserves its degree $d \geq 2$. The proof fixes the polarization isomorphism and its section identities, chooses base-point-free forms over a finite separable extension, and descends all their conjugate tuples together.

Theorem 3.1. *Let k be an arbitrary field, let X be a nonempty projective k -scheme, let $\mathcal{L}$ be an ample line bundle, and suppose that $\phi : X \rightarrow X$ satisfies (3.1) with $d \geq 2$. There exist integers $s \geq 1$, $M \geq 1$, a closed immersion $j : X \hookrightarrow \mathbb{P}_k^M$, and a morphism $\Psi : \mathbb{P}_k^M \rightarrow \mathbb{P}_k^M$ such that*

$$j^* \mathcal{O}_{\mathbb{P}^M}(1) \simeq \mathcal{L}^{\otimes s}, \quad \Psi^* \mathcal{O}_{\mathbb{P}^M}(1) \simeq \mathcal{O}_{\mathbb{P}^M}(d), \quad \Psi j = j \phi. \quad (3.3)$$

In particular, Ψ has algebraic degree d , and $\Psi^m j = j \phi^m$ for every integer $m \geq 0$.

The scheme may be nonreduced, reducible, or nonequidimensional, and the embedding may be linearly degenerate. The ambient dimension and s depend on the given data. For the empty scheme, the conclusion follows from the map $[u : v] \mapsto [u^d : v^d]$ on $\mathbb{P}_k^1$. Fix a k -isomorphism

$$\mu : \phi^* \mathcal{L} \xrightarrow{\sim} \mathcal{L}^{\otimes d}, \quad \mu_s = \mu^{\otimes s}. \quad (3.4)$$

All pullbacks of sections will be identified through this isomorphism and its tensor powers.

3.1 Lifting sections and eliminating geometric base points

We first choose coordinates in which the pulled-back sections lift to forms of degree d . We may then change the lifts by adding forms that vanish on X . The next lemma also ensures that these changes can affect the coordinate values at every geometric point outside X .

Lemma 3.2. *There exist $s \geq 1$, $n \geq 1$, and sections $s_0, \dots, s_n \in H^0(X, \mathcal{L}^{\otimes s})$ with the following properties.*

1. *The sections define a closed immersion $j_0 : X \hookrightarrow \mathbb{P}_k^n$.*
2. *Each $\mu_s(\phi^* s_i)$ is a homogeneous polynomial of degree d in $s_0, \dots, s_n$, with coefficients in k .*
3. *Put*

$$V = \ker \left(H^0(\mathbb{P}_k^n, \mathcal{O}(d)) \longrightarrow H^0(X, \mathcal{L}^{\otimes sd}) \right).$$

For every geometric point $P \in \mathbb{P}_k^n \setminus j_0(X_{\bar{k}})$, the evaluation map $V \otimes_k \bar{k} \rightarrow \mathcal{O}(d)|_P$ is nonzero.

Proof. Choose $a \geq 1$ such that $\mathcal{B} = \mathcal{L}^{\otimes a}$ is very ample, and choose an associated closed immersion $i : X \hookrightarrow \mathbb{P}_k^r$. By adding a zero coordinate, we may assume $r \geq 1$. Let $I \subset k[u_0, \dots, u_r]$ be the homogeneous ideal of the embedding, with homogeneous generators $g_1, \dots, g_h$ of degrees $c_1, \dots, c_h$. Let $\mathcal{I}$ be the corresponding ideal sheaf. For $m \geq 0$, write $\mathcal{I}(m) = \mathcal{I} \otimes_{\mathbb{P}^r}(m)$. The exact sequence

$$0 \longrightarrow \mathcal{I}(m) \longrightarrow \mathcal{O}_{\mathbb{P}^r}(m) \longrightarrow i_* \mathcal{B}^{\otimes m} \longrightarrow 0$$

shows that $H^1(\mathbb{P}^r, \mathcal{I}(m)) = 0$ is sufficient for lifting sections; H^1 denotes the first sheaf cohomology group. Serre vanishing gives this condition for all sufficiently large m . Choose $b \geq 1$ large enough that $b \geq c_j$ for every j and both maps

$$\begin{aligned} H^0(\mathbb{P}^r, \mathcal{O}(b)) &\longrightarrow H^0(X, \mathcal{B}^{\otimes b}), \\ H^0(\mathbb{P}^r, \mathcal{O}(bd)) &\longrightarrow H^0(X, \mathcal{B}^{\otimes bd}) \end{aligned} \quad \text{are surjective.} \quad (3.5)$$

Serre vanishing is applied to the same fixed ideal sheaf [11, Tag 0B5T].

Let $\nu_b : \mathbb{P}^r \hookrightarrow \mathbb{P}^n$ be the degree- b Veronese embedding, whose coordinates are all monomials of degree b ; thus $n+1 = \binom{r+b}{b}$. For example, when $r = 1$ and $b = 2$, $\nu_2[u_0 : u_1] = [u_0^2 : u_0u_1 : u_1^2]$, with image the conic $Y_0Y_2 = Y_1^2$. In general, the Veronese embedding similarly treats higher-degree monomials in the original coordinates as new coordinates. Put $s = ab$, and let $s_0, \dots, s_n$ be the restrictions of these monomials to X . Then $j_0 = \nu_b i$ is a closed immersion. Every degree- bd monomial in the original coordinates factors into d monomials of degree b . Hence the space of degree- d forms in the new coordinates surjects onto the space of degree- bd forms in the original coordinates. Together with (3.5), this proves the second assertion.

We next verify that the ideal of the Veronese image is generated by quadrics. Write Y_α for the new coordinate corresponding to u^α , where $\alpha = (\alpha_0, \dots, \alpha_r) \in \mathbb{N}_0^{r+1}$, $|\alpha| = \sum_i \alpha_i = b$, and $u^\alpha = \prod_i u_i^{\alpha_i}$. The kernel of the graded homomorphism

$$k[Y_\alpha : |\alpha| = b] \longrightarrow \bigoplus_{m \geq 0} k[u_0, \dots, u_r]_{mb}, \quad Y_\alpha \longmapsto u^\alpha$$

is spanned by differences of monomials with the same image. For such a pair, write the multi-index of each factor as a row. The two arrays have equal column sums, and each row has sum b . If the first row differs from its target, choose an excess coordinate i and a deficient coordinate j . Some other row has a positive entry in coordinate j . Exchanging one unit in coordinates i and j between that row and the first gives the relation

$$Y_\alpha Y_\beta - Y_{\alpha - e_i + e_j} Y_{\beta + e_i - e_j},$$

where e_i, e_j are standard unit vectors. This exchange reduces the discrepancy in the first row, preserves nonnegative entries, and preserves every row sum. Once the first row agrees with its target, induction on the remaining rows connects the two monomials. Thus these quadratic relations generate the whole kernel, in every characteristic.

Now take P as in the third assertion. If $P \notin \nu_b(\mathbb{P}_k^r)$, some Veronese quadratic relation is nonzero at P . If $P = \nu_b(Q)$ with $Q \notin i(X_{\bar{k}})$, then $g_j(Q) \neq 0$ for some j . Choose a coordinate linear form ℓ nonzero at Q . The degree- b polynomial $g_j \ell^{b-c_j}$ vanishes on X and is nonzero at Q ; it corresponds to a linear form in the new coordinates. In either case, we obtain a form of degree 2 or 1 that vanishes as a section on $j_0(X_{\bar{k}})$ and is nonzero at P . Multiplying by a suitable power of a new coordinate linear form nonzero at P raises its degree to $d \geq 2$. Flatness of field extension identifies the kernel of the base-changed restriction map with $V \otimes_k \bar{k}$, proving the third assertion. $\square$

Put $\mathcal{M} = \mathcal{L}^{\otimes s}$. By Lemma 3.2, choose degree- d forms $F_0^0, \dots, F_n^0$ over k such that

$$F_i^0(s_0, \dots, s_n) = \mu_s(\phi^* s_i) \quad \text{in } H^0(X, \mathcal{M}^{\otimes d}). \quad (3.6)$$

All lifts satisfying these identities form the affine space

$$\mathcal{P} = \prod_{i=0}^n (F_i^0 + V). \quad (3.7)$$

Lemma 3.3. *The tuples in $\mathcal{P}$ with no common geometric base point form a nonempty open k -subscheme $\mathcal{U}$. If k is infinite, then $\mathcal{U}(k) \neq \emptyset$. If k is finite, there is a finite separable extension K/k with $\mathcal{U}(K) \neq \emptyset$.*

Proof. Consider the common-zero closed subscheme of $\mathcal{P} \times \mathbb{P}^n$ given by

$$\mathcal{Z} = \{((F_0, \dots, F_n), P) : F_0(P) = \dots = F_n(P) = 0\}.$$

The sections $s_0, \dots, s_n$ generate $\mathcal{M}$, and their pullbacks, identified through μ_s , generate $\mathcal{M}^{\otimes d}$. By (3.6), $\mathcal{Z}$ is empty over $\mathcal{P} \times j_0(X)$. For a geometric point P in the complement, the third assertion of Lemma 3.2 says that each condition $F_i(P) = 0$ defines a nonempty affine hyperplane in the parameter factor $F_i^0 + V$. These conditions involve distinct factors, so together they have codimension $n + 1$. If the complement is nonempty, it follows that

$$\dim \mathcal{Z}_{\bar{k}} \leq \dim \mathcal{P}_{\bar{k}} + n - (n + 1) = \dim \mathcal{P} - 1. \quad (3.8)$$

If the complement is empty, then $\mathcal{Z}$ itself is empty.

The projection $\mathcal{P} \times \mathbb{P}^n \rightarrow \mathcal{P}$ is projective, so the image of $\mathcal{Z}$ is closed. The dimension bound makes it a proper closed subset. Its complement is the required nonempty open set $\mathcal{U}$. Over an infinite field, the k -points of affine space are Zariski dense, so $\mathcal{U}$ has a k -point. Over a finite field, the nonempty finite-type k -scheme $\mathcal{U}$ has a closed point whose residue field K is finite over k . Every finite extension of a finite field is separable. $\square$

3.2 Homogeneous descent along a finite separable extension

A nonempty open set over a finite field can have no rational points. For example, $\mathbb{A}_{\mathbb{F}_q}^1 \setminus \{x : x^q - x = 0\}$ is nonempty and has no $\mathbb{F}_q$ -points. Lemma 3.3 chooses the required lifts over a finite extension. We now expand each extension-field coordinate in a basis over the base field. Separability splits the resulting equations over $\bar{k}$ into independent conjugate tuples, preserving the absence of geometric base points.

Lemma 3.4. *Let K/k be a finite separable extension of degree ρ , and let $N \geq 1$. Let $F = (F_1, \dots, F_N)$ be a tuple of degree- d homogeneous polynomials over K whose only common zero in $\mathbb{A}_{\bar{k}}^N$ is the origin. Regard K^N as a k -vector space W of dimension ρN , and expand F in a k -basis to obtain a polynomial map $G : W \rightarrow W$ over k . The coordinates of G are homogeneous of degree d , and their only common zero in $W_{\bar{k}}$ is the origin. Consequently, if $\rho N \geq 2$, then G defines a degree- d morphism $\mathbb{P}(W) \rightarrow \mathbb{P}(W)$.*

Proof. Choose a basis $\beta_1, \dots, \beta_\rho$ of K/k and introduce k -coordinates x_{ia} for $1 \leq i \leq N$, $1 \leq a \leq \rho$. Using the multiplication structure constants of K in this basis, write uniquely

$$F_i \left(\sum_a \beta_a x_{1a}, \dots, \sum_a \beta_a x_{Na} \right) = \sum_a \beta_a G_{ia}(\mathbf{x}). \quad (3.9)$$

Each G_{ia} is a degree- d form over k .

Let Σ be the set of k -embeddings $K \hookrightarrow \bar{k}$. Separability gives

$$K \otimes_k \bar{k} \xrightarrow{\sim} \prod_{\sigma \in \Sigma} \bar{k}, \quad b \otimes c \longmapsto (\sigma(b)c)_\sigma. \quad (3.10)$$

Hence the linear change of variables

$$z_i^\sigma = \sum_a \sigma(\beta_a) x_{ia} \quad (3.11)$$

is invertible. Applying each σ to (3.9) gives

$$\sum_a \sigma(\beta_a) G_{ia}(\mathbf{x}) = F_i^\sigma(z_1^\sigma, \dots, z_N^\sigma). \quad (3.12)$$

Under this same change of variables in the source and target, $G_{\bar{k}}$ becomes the direct sum of all F^σ . Each k -embedding σ extends to a k -automorphism of $\bar{k}$, so each conjugate tuple F^σ also has only the origin as a common geometric zero. If $G(\mathbf{x}) = 0$, then (3.12) gives $z_i^\sigma = 0$ for every i, σ . The invertibility of (3.11) yields $\mathbf{x} = 0$. Thus the homogeneous coordinates have no common zero on $\mathbb{P}(W)$ and define the asserted morphism, with pullback of $\mathcal{O}(1)$ equal to $\mathcal{O}(d)$. $\square$

The construction restricts scalars on an affine vector space and then projectivizes once. Fakhruddin used restriction of scalars for projective schemes in arithmetic dynamics over number fields, followed by a further extension using the infinite-field theorem [6, Remark 2.6 of the preprint]. Lemma 3.4 directly establishes the absence of geometric base points in homogeneous coordinates. Its ambient space is $\mathbb{P}(W)$, of dimension $\rho N - 1$.

An example over a quadratic extension. Take $k = \mathbb{F}_3$ and $K = k(\alpha)$ with $\alpha^2 = -1$. Expanding the square of $z = x + \alpha y$ in the basis $1, \alpha$ gives

$$z^2 = (x^2 - y^2) + \alpha(2xy), \quad G(x, y) = (x^2 - y^2, 2xy).$$

Over $\bar{k}$, the equations $x^2 - y^2 = 2xy = 0$ force $x = y = 0$. Thus G projectivizes to a quadratic morphism of $\mathbb{P}_{\bar{k}}^1$. The two conjugate coordinates $x + \alpha y$ and $x - \alpha y$ are each squared, giving the two tuples in (3.12).

Example 3.5. Separability is necessary in Lemma 3.4. Suppose $\text{char } k = 2$, choose $a \in k \setminus k^2$, and let $K = k(\alpha)$ with $\alpha^2 = a$. The tuple $F(u, v) = (u^2, v^2)$ has only the origin as a common zero in $\mathbb{A}_{\bar{k}}^2$. Substituting $u = x_1 + \alpha y_1$, $v = x_2 + \alpha y_2$ gives the restriction-of-scalars map

$$G(x_1, y_1, x_2, y_2) = (x_1^2 + ay_1^2, 0, x_2^2 + ay_2^2, 0).$$

It vanishes at the nonzero geometric vector $(\alpha, 1, 0, 0)$, so the corresponding homogeneous forms have a base point on $\mathbb{P}_{\bar{k}}^3$.

Proof of Theorem 3.1. Choose the sections of Lemma 3.2 and the open set $\mathcal{U}$ of Lemma 3.3. If k is infinite, choose $(F_0, \dots, F_n) \in \mathcal{U}(k)$ and put $j = j_0$, $\Psi = [F_0 : \dots : F_n]$. The absence of common geometric base points makes Ψ a morphism, and (3.6) gives $\Psi j = j\phi$.

Suppose that k is finite. Choose a finite separable extension K/k and $(F_0, \dots, F_n) \in \mathcal{U}(K)$, and put $\rho = [K : k]$. These lifts satisfy the section identities on the original scheme

$$F_i(s_0, \dots, s_n) = \mu_s(\phi^* s_i) \quad \text{in } H^0(X, \mathcal{M}^{\otimes d}) \otimes_k K. \quad (3.13)$$

Here global sections commute with field extension. For example, the identification follows by computing sections as the equalizer for a finite affine open cover and using the flatness of K/k .

Choose a k -basis $\beta_1, \dots, \beta_\rho$ with $\beta_1 = 1$, and apply Lemma 3.4 to F . Let W be the underlying k -vector space of K^{n+1} . We obtain $M = \rho(n+1) - 1 \geq 1$ and a degree- d morphism $\Psi : \mathbb{P}(W) \rightarrow \mathbb{P}(W)$, with coordinates G_{ia} indexed by $0 \leq i \leq n$, $1 \leq a \leq \rho$. Define $j : X \hookrightarrow \mathbb{P}(W)$ by the sections

$$(s_0, 0, \dots, 0; s_1, 0, \dots, 0; \dots; s_n, 0, \dots, 0). \quad (3.14)$$

This is the composition of j_0 with a linear closed immersion, and $j^* \mathcal{O}(1) = \mathcal{M}$.

Substitute (3.14) into (3.9) and use (3.13) to obtain

$$\sum_a \beta_a G_{ia}(s_0, 0, \dots, s_n, 0) = \mu_s(\phi^* s_i).$$

The right-hand side is a section defined over k . Comparing coefficients in the k -basis $1, \beta_2, \dots, \beta_\rho$ gives

$$\begin{aligned} G_{i1}(s_0, 0, \dots, s_n, 0) &= \mu_s(\phi^* s_i), \\ G_{ia}(s_0, 0, \dots, s_n, 0) &= 0 \quad (a > 1). \end{aligned} \quad (3.15)$$

These identities hold in $H^0(X, \mathcal{M}^{\otimes d})$. The generating tuples of sections of the same line bundle therefore give the equality of scheme morphisms $\Psi j = j\phi$. Comparing coefficients of sections retains the nonreduced structure of X . Induction on m now yields $\Psi^m j = j\phi^m$. $\square$

Remark 3.6. In the finite-field construction, the embedded image lies in an n -dimensional linear subspace. Outside X , this subspace need not be preserved by Ψ : the additional coordinates participate in the common base-point-free condition for all conjugate tuples in (3.12). The theorem allows the ambient dimension to increase while retaining a tensor power of the original polarization line bundle and every iterate of the original map.

3.3 The degree-one boundary

Poonen's question on changing the embedding also includes $d = 1$. For a general projective scheme, this boundary depends on whether ϕ is an automorphism. In this subsection, the existence of an extension allows any new embedding and any ambient degree.

Proposition 3.7. *Let X be a nonempty projective k -scheme, let $\mathcal{L}$ be ample, and suppose that $\phi^*\mathcal{L} \simeq \mathcal{L}$. If $\dim X = 0$, then ϕ extends to an ambient morphism for some projective embedding. If $\dim X > 0$, such an extension exists if and only if ϕ is an automorphism. In that case an extension of degree one exists, and every extension has degree one.*

Proof. When $\dim X = 0$, the scheme X is finite over k and hence affine, so $\mathcal{O}_X$ is ample. Applying Theorem 3.1 to $\phi^*\mathcal{O}_X \simeq \mathcal{O}_X^{\otimes 2}$ gives an extension of degree two.

Suppose that $\dim X > 0$. First, ϕ is finite. Indeed, ϕ is proper, and the restriction of $\phi^*\mathcal{L}$ to any geometric fiber is trivial. A positive-dimensional geometric fiber would contain an integral projective curve. On that curve, $\mathcal{L} \simeq \phi^*\mathcal{L}$ would have both positive and zero degree, a contradiction. Thus every fiber has dimension zero. Properness and quasi-finiteness imply finiteness [11, Tag 02LS].

Suppose that $j : X \hookrightarrow \mathbb{P}^M$ and Ψ give an extension, and put $\mathcal{A} = j^*\mathcal{O}(1)$. Since ϕ is finite and X has positive dimension, Ψ is nonconstant. There is therefore an integer $e \geq 1$ such that

$$\Psi^*\mathcal{O}(1) \simeq \mathcal{O}(e), \quad \phi^*\mathcal{A} \simeq \mathcal{A}^{\otimes e}. \quad (3.16)$$

Choose b large enough that $\mathcal{L}^{\otimes b} \otimes \mathcal{A}^{-1}$ is ample. Pullback by a finite morphism preserves ampleness [11, Tag 0892], so for every $m \geq 0$ the line bundle

$$(\phi^m)^*(\mathcal{L}^{\otimes b} \otimes \mathcal{A}^{-1}) \simeq \mathcal{L}^{\otimes b} \otimes \mathcal{A}^{\otimes(-e^m)} \quad (3.17)$$

is ample. Fix an integral projective curve C in $X_{\bar{k}}$. Then

$$b \deg_C \mathcal{L} - e^m \deg_C \mathcal{A} > 0 \quad \text{for every } m \geq 0.$$

Both degrees are positive, and hence $e = 1$.

An everywhere-defined degree-one morphism of projective space is given by an invertible matrix, so Ψ is an automorphism. Put $Z = \Psi(j(X))$. Then $Z \subseteq j(X)$ is an inclusion of closed subschemes with equal Hilbert polynomials relative to $\mathcal{O}(1)$. A strict inclusion would give a nonzero coherent ideal sheaf $\mathcal{I}$ with zero Hilbert polynomial. For sufficiently large m , the twist $\mathcal{I}(m)$ is globally generated and has vanishing higher cohomology. Since it is nonzero, it has a nonzero global section, contradicting the zero Hilbert polynomial. Thus $Z = j(X)$, and the restriction of Ψ to this closed subscheme is an automorphism. The extension identity makes ϕ an automorphism as well.

Conversely, suppose that ϕ is an automorphism and choose a very ample power $\mathcal{L}^{\otimes s}$. Through $\phi^*\mathcal{L}^{\otimes s} \simeq \mathcal{L}^{\otimes s}$, pullback of sections gives an invertible k -linear operator on $H^0(X, \mathcal{L}^{\otimes s})$. After choosing a basis, its matrix gives a degree-one extension for the embedding by the complete linear system. $\square$

Example 3.8. Over any field, take the connected reduced curve

$$X = \{UV = 0\} \subset \mathbb{P}_k^2, \quad \phi[U : V : Z] = [U + V : 0 : Z], \quad \mathcal{L} = \mathcal{O}_X(1).$$

The equations $U + V = Z = UV = 0$ imply $U = V = Z = 0$, so the sections have no geometric base point on X . They define a morphism $\phi : X \rightarrow X$ with $\phi^*\mathcal{L} \simeq \mathcal{L}$. Both lines map to $\{V = 0\}$, and $[1 : 0 : 0]$ and $[0 : 1 : 0]$ have the same image. Thus ϕ is not an automorphism. Proposition 3.7 rules out an ambient morphism extending ϕ after any change of embedding.

Remark 3.9. If X is an integral projective variety and $\phi^*\mathcal{L} \simeq \mathcal{L}$, the finiteness above implies that the image of ϕ has the same dimension as X and is therefore dense. Integrality makes pullback of line-bundle sections injective, so its linear action on $H^0(X, \mathcal{L}^{\otimes s})$ is invertible. The complete linear system gives a linear extension, and Proposition 3.7 then makes ϕ an automorphism. This degree-one case for integral varieties is already contained in Fakhruddin's corollary. Poonen's introduction discusses arbitrary projective schemes when citing the infinite-field result [5, §1]. Example 3.8 gives a negative instance of the change-of-embedding question for reducible schemes when $d = 1$, and Proposition 3.7 gives the extension criterion in this range.

4 Orbit-density oscillation over finite fields

Let $f : \mathbb{P}^N \dashrightarrow \mathbb{P}^N$ be a dominant rational map defined over a field k , so that the Zariski closure of its image is all of $\mathbb{P}^N$. The indeterminacy locus $I(f)$ consists of the points at which f cannot be defined regularly on a neighborhood. In homogeneous coordinates on projective space, after canceling the common factors of the homogeneous polynomials representing f , their common zero locus is $I(f)$. We define forward orbits by applying the original map one step at a time: starting from $P_0 = P$, set $P_{j+1} = f(P_j)$ whenever $P_j \notin I(f)$. If some P_j lies in $I(f)$, the subsequent orbit is undefined.

Definition 4.1. For a field extension L/k , write

$$\mathbb{P}^N(L)_f = \{P \in \mathbb{P}^N(L) : P_j \text{ is defined for every } j \geq 0 \text{ and satisfies } P_j \notin I(f)\}. \quad (4.1)$$

This is the locus of points whose forward orbit is defined at every step, which we call the *everywhere-defined orbit locus* of f over L . Equivalently, on L -points set $D_0 = \mathbb{P}^N(L) \setminus I(f)$ and define $D_{m+1} = \{P \in D_0 : f(P) \in D_m\}$ recursively. Then $\mathbb{P}^N(L)_f = \bigcap_{m \geq 0} D_m$.

This agrees with the notation in the survey of Benedetto et al. [8, §3, Conjecture 18.10(b)]. Here the density over a finite field is the proportion of rational points whose forward orbit is defined at every step. Fix a finite field $\mathbb{F}_Q$ and let n denote the extension degree, so that $\mathbb{F}_{Q^n}$ has Q^n elements. The density conjecture in that survey asserts that every dominant rational self-map defined over $\mathbb{F}_Q$ satisfies

$$\lim_{n \rightarrow \infty} \frac{\#\mathbb{P}^N(\mathbb{F}_{Q^n})_f}{\#\mathbb{P}^N(\mathbb{F}_{Q^n})} = 1. \quad (4.2)$$

Weinreich questioned this assertion in his study of the pentagram map and conjectured that the density of its everywhere-defined orbit locus tends to zero over finite-field extensions [9, §8, Conjecture 8.3]. We give a cubic birational map of the projective plane whose orbit density fails to converge over the extensions of every fixed finite base field. Over $\mathbb{F}_2$, its upper and lower limits are 1 and 0, respectively.

The construction turns returns in a finite multiplicative group into visits to the indeterminacy locus, making the number of points in the everywhere-defined orbit locus depend on the orders

of the group elements. In homogeneous coordinates $[A : B : Z]$ on the projective plane, define the homogeneous quadratic polynomial and rational map

$$\begin{aligned} R(A, B, Z) &= B(A + Z) - AZ, \\ f[A : B : Z] &= [AR(A, B, Z) : BR(A, B, Z) : A^2Z]. \end{aligned} \quad (4.3)$$

These formulas have integer coefficients and define the same construction in every characteristic. For a positive integer m , let $\varphi(m) = \#\{1 \leq j \leq m : \gcd(j, m) = 1\}$ be Euler's totient function, and set

$$\mathcal{A}(m) = \sum_{d|m} d\varphi(d), \quad \mathcal{M}(m) = \frac{\mathcal{A}(m)}{m^2}. \quad (4.4)$$

The order of an element x of a finite group is the least positive integer d for which $x^d = 1$. A cyclic group with m elements has $\varphi(d)$ elements of order exactly $d \mid m$. Thus $\mathcal{A}(m)$ is the sum of all element orders, $\mathcal{A}(m)/m$ is the mean element order, and $\mathcal{M}(m)$ is this mean normalized by the group order m . In particular, $\mathcal{A}(1) = \mathcal{M}(1) = 1$.

Theorem 4.2. *The map (4.3) is a cubic birational map over every finite field, with indeterminacy locus*

$$I(f) = \{O, E, F\}, \quad O = [0 : 0 : 1], \quad E = [0 : 1 : 0], \quad F = [1 : 0 : 0]. \quad (4.5)$$

For every finite field $\mathbb{F}_q$,

$$\#\mathbb{P}^2(\mathbb{F}_q)_f = q^2 + q - 2 - \mathcal{A}(q - 1). \quad (4.6)$$

For any fixed finite base field $\mathbb{F}_Q$, let

$$\delta_Q(n) = \frac{\#\mathbb{P}^2(\mathbb{F}_{Q^n})_f}{Q^{2n} + Q^n + 1}. \quad (4.7)$$

Then

$$\liminf_{n \rightarrow \infty} \delta_Q(n) = 1 - \mathcal{M}(Q - 1), \quad \limsup_{n \rightarrow \infty} \delta_Q(n) = 1. \quad (4.8)$$

Consequently, $\delta_Q(n)$ fails to converge for every fixed finite base field. When $Q = 2$, its lower limit is 0 and its upper limit is 1.

4.1 The indeterminacy locus and coordinates for the original orbit

Lemma 4.3. *The three homogeneous coordinates in (4.3) have no nonconstant common factor. The indeterminacy locus of the map is exactly (4.5). The map admits no regular extension at any of these three points, in any characteristic.*

Proof. In the polynomial ring over any field, A and B are coprime, so $\gcd(AR, BR) = R$ up to a nonzero constant factor. The identities $R(0, B, Z) = BZ$ and $R(A, B, 0) = AB$ show that neither A nor Z divides R . Hence $\gcd(R, A^2Z) = 1$: the three coordinates have no common factor, and each has degree 3.

We determine their common zeros. If $A = 0$, the second coordinate is B^2Z , giving only O and E . If $A \neq 0$, vanishing of the third coordinate gives $Z = 0$, and vanishing of the first then gives $B = 0$, yielding F . These are the three stated base points.

To show that the map has no regular extension at these points, we compare two formal arcs through each one. The arcs have parameter ε , and their generic points take values in the field of formal Laurent series. The image specializes by setting $\varepsilon = 0$ in its homogeneous coordinates after canceling common powers of ε . If f were regular at the common initial point of the two arcs, their images would have to specialize to the same point.

Near O , use the affine coordinates $(a, b) = (A/Z, B/Z)$. Along $(a, b) = (\varepsilon, 0)$, the image is constantly $[-1 : 0 : 1]$; along $(a, b) = (\varepsilon, \varepsilon)$, it is $[\varepsilon : \varepsilon : 1]$, which specializes to O . These two specializations are distinct.

Near E , use the coordinates $(a, z) = (A/B, Z/B)$ on $B = 1$. Along $(a, z) = (\varepsilon, 0)$, the image is $[\varepsilon : 1 : 0]$ and specializes to E . Along

$$(a, z) = \left(\varepsilon, \frac{\varepsilon}{\varepsilon - 1} \right)$$

we have $R = a + z - az = 0$, while $a^2 z \neq 0$ at the generic point, so the image is constantly O . Here $\varepsilon - 1$ is a unit at $\varepsilon = 0$, so both arcs are defined.

Near F , use the coordinates $(b, z) = (B/A, Z/A)$ on $A = 1$. Along $(b, z) = (\varepsilon, 0)$, the image is $[1 : \varepsilon : 0]$ and specializes to F ; along $(b, z) = (0, \varepsilon)$, the image is constantly $[-1 : 0 : 1]$. The two specializations in each pair remain distinct in characteristic 2, so the map has no regular extension at any of the three points. $\square$

On an affine open set, orbit calculations reduce to multiplicative iteration. Consider

$$U = \{[A : B : Z] : AZ \neq 0\}.$$

On $Z = 1$, write $a = A/Z$ and $b = B/Z$. As long as the orbit remains in U , the slope b/a is constant, while the translated coordinate $a + 1$ is multiplied by that slope at each step. This suggests the following coordinate isomorphism:

$$u = \frac{b}{a}, \quad v = a + 1; \quad \Phi(u, v) = [v - 1 : u(v - 1) : 1], \quad v \neq 1. \quad (4.9)$$

The coordinate space is $\mathbb{A}_u^1 \times (\mathbb{A}_v^1 \setminus \{1\})$.

Lemma 4.4. *The map f is birational. For $P = \Phi(u, v) \in U$, the value $f(P)$ is always defined and satisfies*

$$f(P) = [uv - 1 : u(uv - 1) : 1]. \quad (4.10)$$

If $uv \neq 1$, then $f(P) = \Phi(u, uv) \in U$; if $uv = 1$, then $f(P) = O$. Thus, until the original orbit reaches O , its successive coordinates satisfy

$$u_j = u, \quad v_j = u^j v, \quad P_j = [u^j v - 1 : u(u^j v - 1) : 1]. \quad (4.11)$$

The last identity remains valid at the first time when $u^j v = 1$; at that time, $P_j = O$.

Proof. When $a \neq 0$, the third coordinate a^2 in (4.3) is nonzero, so direct normalization gives

$$a' = \frac{b(a + 1) - a}{a} = uv - 1, \quad b' = \frac{b}{a} \frac{b(a + 1) - a}{a} = u(uv - 1). \quad (4.12)$$

This step is defined even when $uv = 1$, in which case its image is O . When $uv \neq 1$, the coordinates of the next point are $u' = b'/a' = u$ and $v' = a' + 1 = uv$. Hence, on dense open sets, f is conjugate to

$$S(u, v) = (u, uv) \quad (4.13)$$

On $u \neq 0$, the latter has rational inverse $(u, v) \mapsto (u, v/u)$. Composing with (4.9) gives a rational inverse of f . The conjugacy is given by a coordinate isomorphism between dense open subvarieties; the rational inverse formulas hold even over finite base fields with very few rational points. Thus f is birational, and in particular dominant, in every characteristic.

Applying (4.12) to the original orbit one step at a time proves (4.11) by induction. If $j \geq 1$ is the first time such that $u^j v = 1$, then $P_0, \dots, P_{j-1}$ all lie in U , and $f(P_{j-1})$ is defined and equals $O \in I(f)$. The next step $f(O)$ is undefined, so iteration stops there. $\square$

Over $\mathbb{F}_5$, take $u = 2$ and $v = 2$. The element 2 has multiplicative order 4, and the successive values of v_j are 2, 4, 3, 1. The corresponding original orbit is

$$[1 : 2 : 1] \mapsto [3 : 1 : 1] \mapsto [2 : 4 : 1] \mapsto O.$$

The first three steps are defined; the third reaches O , and the fourth is undefined. The triangular map S still sends $(2, 1)$ to $(2, 2)$, but this step no longer corresponds to the orbit of the original map f .

In general, the coordinate formula (4.11) applies through the first arrival at O . At later times its coordinate expressions no longer describe the original orbit. By Definition 4.1, every initial point that reaches O is excluded from the everywhere-defined orbit locus. Through the first arrival at O , each composition is regular near the initial point. Thus, even when f^j is viewed as the rational composite after cancellation, its value at this initial point is still O .

4.2 Counting all finite-field points

Proposition 4.5. *For every finite field $\mathbb{F}_q$, exactly $\mathcal{A}(q-1) + 3$ points have a forward orbit that fails to be defined at some step. Consequently, (4.6) holds.*

Proof. Partition the projective plane into the three disjoint parts $Z = 1, a \neq 0$, $Z = 1, a = 0$, and $Z = 0$. The first part is in bijection with

$$(u, v) \in \mathbb{F}_q \times (\mathbb{F}_q \setminus \{1\})$$

If $u = 0$, formula (4.10) sends the point in one step to $[-1 : 0 : 1]$, a fixed point at which f is defined. If $v = 0$, the initial point is already fixed. Both types of initial points belong to the everywhere-defined orbit locus.

Now suppose $u, v \neq 0$ and $v \neq 1$. By Lemma 4.4, the original orbit fails to be defined at some step if and only if $u^j v = 1$ for some positive integer j . Let $d = \text{ord}(u)$ be the order of u in the multiplicative group, namely the least positive integer for which $u^d = 1$, and let $\langle u \rangle = \{1, u, \dots, u^{d-1}\}$ be the cyclic subgroup generated by u . The condition is equivalent to

$$v \in \langle u \rangle \setminus \{1\}. \quad (4.14)$$

Indeed, for a point in this set, choose the least positive j . All earlier steps stay in U , and step j reaches O . Conversely, if $v \notin \langle u \rangle$, then all $u^j v - 1$ are nonzero, so the entire orbit remains in U . Thus, for each fixed $u \neq 0$, exactly $d - 1$ initial values of v yield an orbit that becomes undefined.

The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $m = q - 1$ and has $\varphi(d)$ elements of order $d \mid m$. The number of excluded points in the first part is therefore

$$\sum_{u \in \mathbb{F}_q^\times} (\text{ord}(u) - 1) = \sum_{d \mid m} (d - 1) \varphi(d) = \mathcal{A}(m) - m. \quad (4.15)$$

Here we used $\sum_{d \mid m} \varphi(d) = m$, which also follows by partitioning the cyclic group according to element order.

The second part consists of the q points on the affine line $a = 0$. The point with $b = 0$ is O ; the original homogeneous formulas send each of the other points to E in one step. All points in this part are therefore excluded.

The third part is the line at infinity. The two coordinate points E, F lie in $I(f)$; the remaining $q - 1$ points satisfy $AB \neq 0$ and

$$f[A : B : 0] = [A^2 B : AB^2 : 0] = [A : B : 0], \quad (4.16)$$

so they are fixed points at which f is defined. The total number of excluded points in the three disjoint parts is

$$(\mathcal{A}(m) - m) + q + 2 = \mathcal{A}(m) + 3.$$

Subtracting this from $\#\mathbb{P}^2(\mathbb{F}_q) = q^2 + q + 1$ proves the result. $\square$

This count includes the special affine values $u = 0$ and $v = 0$, the entire boundary line $a = 0$, and the line at infinity, so it also applies to the smallest finite field $\mathbb{F}_2$. Condition (4.14) counts each initial point that reaches O at any time exactly once. For example, in $\mathbb{F}_5^\times$, the elements 1, 2, 3, 4 have orders 1, 4, 4, 2, respectively, so $\mathcal{A}(4) = 11$. Of the 31 points of the projective plane, $11 + 3 = 14$ are excluded, and the remaining 17 belong to the everywhere-defined orbit locus. Its density is $17/31$.

4.3 Mean element order and extension degrees

The counting formula reduces the density problem to the behavior of $\mathcal{M}(Q^n - 1)$. Two types of extension degree lead to different arithmetic behavior. As n runs through increasingly large primes, the prime divisors of $Q^n - 1$ outside the fixed factor $Q - 1$ grow larger, and their total effect on the normalized mean element order tends to zero. When n is a factorial, $Q^n - 1$ is divisible by increasingly many small primes, forcing this normalized mean to tend to zero. We first establish a product formula for $\mathcal{M}$ and then prove these two assertions.

Lemma 4.6. *The functions $\mathcal{A}$ and $\mathcal{M}$ are multiplicative: each sends a product of coprime integers to the product of its values. For every positive integer m ,*

$$\mathcal{M}(m) = \prod_{r^e \parallel m} \left(\frac{r}{r+1} + \frac{1}{(r+1)r^{2e}} \right), \quad (4.17)$$

where r ranges over primes and $r^e \parallel m$ means $r^e \mid m$ but $r^{e+1} \nmid m$. For positive integers d, m ,

$$d \mid m \implies \mathcal{M}(m) \leq \mathcal{M}(d), \quad (4.18)$$

and

$$\frac{\varphi(m)}{m} \leq \mathcal{M}(m) \leq 1. \quad (4.19)$$

Proof. Since $d \mapsto d\varphi(d)$ is multiplicative, its divisor sum $\mathcal{A}(m)$ is also multiplicative. Dividing by m^2 gives the multiplicativity of $\mathcal{M}$. For a prime power,

$$\mathcal{A}(r^e) = 1 + \sum_{j=1}^e (r^{2j} - r^{2j-1}) = \frac{r^{2e+1} + 1}{r + 1}.$$

Division by r^{2e} gives (4.17). Each local factor is 1 at $e = 0$ and decreases as e increases, which proves (4.18). Finally, let C_m be a cyclic group of order m . Every element has order at most m , and there are $\varphi(m)$ generators of order m . Hence

$$m\varphi(m) \leq \sum_{x \in C_m} \text{ord}(x) = \mathcal{A}(m) \leq m^2.$$

This is (4.19). $\square$

For fixed Q and extension degree n , set $m = Q^n - 1$. By Proposition 4.5, the density can be written as

$$\begin{aligned} \delta_Q(n) &= 1 - \frac{m^2 \mathcal{M}(m) + 3}{m^2 + 3m + 3}, \\ \delta_Q(n) - (1 - \mathcal{M}(m)) &= \frac{(3m + 3)\mathcal{M}(m) - 3}{m^2 + 3m + 3}. \end{aligned} \quad (4.20)$$

Since $0 < \mathcal{M}(m) \leq 1$, the absolute value of the second line is at most $3/m$. Thus

$$\delta_Q(n) = 1 - \mathcal{M}(Q^n - 1) + O((Q^n - 1)^{-1}), \quad (4.21)$$

with an absolute implied constant.

Proposition 4.7. *For every fixed finite-field size Q ,*

$$\liminf_{n \rightarrow \infty} \delta_Q(n) = 1 - \mathcal{M}(Q - 1).$$

The lower limit is attained along prime extension degrees tending to infinity.

Proof. The divisibility $Q - 1 \mid Q^n - 1$ and (4.18) give $\mathcal{M}(Q^n - 1) \leq \mathcal{M}(Q - 1)$. Together with (4.21), this yields

$$\liminf_{n \rightarrow \infty} \delta_Q(n) \geq 1 - \mathcal{M}(Q - 1). \quad (4.22)$$

Let ℓ run through odd primes tending to infinity, omitting the finitely many primes dividing the fixed integer $Q - 1$. Write

$$Q^\ell - 1 = (Q - 1)H_\ell, \quad H_\ell = 1 + Q + \cdots + Q^{\ell-1}. \quad (4.23)$$

Since $H_\ell \equiv \ell \pmod{Q - 1}$, the two factors are coprime. If a prime $r \mid H_\ell$, then $r \nmid Q$. Let $\text{ord}_r(Q)$ be the order of Q in $(\mathbb{Z}/r\mathbb{Z})^\times$. The congruence $Q^\ell \equiv 1 \pmod{r}$ implies that this order divides ℓ . The order cannot be 1, since that would give $r \mid Q - 1$, contradicting coprimality. Hence $\text{ord}_r(Q) = \ell$, so $\ell \mid r - 1$. As ℓ is an odd prime, r is also odd, and

$$r \geq 2\ell + 1. \quad (4.24)$$

Let $\omega(H_\ell)$ denote the number of distinct prime divisors of H_ℓ . Their product is at most H_ℓ , so (4.24) gives

$$\omega(H_\ell) \log(2\ell + 1) \leq \log H_\ell.$$

Consequently,

$$\sum_{r \mid H_\ell} \frac{1}{r} \leq \frac{\log H_\ell}{(2\ell + 1) \log(2\ell + 1)} < \frac{\ell \log Q}{(2\ell + 1) \log(2\ell + 1)} \rightarrow 0. \quad (4.25)$$

The finite-product inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ holds for $0 \leq x_i \leq 1$. Therefore

$$\frac{\varphi(H_\ell)}{H_\ell} = \prod_{r \mid H_\ell} \left(1 - \frac{1}{r}\right) \geq 1 - \sum_{r \mid H_\ell} \frac{1}{r} \rightarrow 1.$$

By (4.19), $\mathcal{M}(H_\ell) \rightarrow 1$. Coprimality and multiplicativity now give

$$\mathcal{M}(Q^\ell - 1) = \mathcal{M}(Q - 1)\mathcal{M}(H_\ell) \rightarrow \mathcal{M}(Q - 1).$$

Thus (4.21) yields

$$\lim_{\substack{\ell \rightarrow \infty \\ \ell \text{ prime}}} \delta_Q(\ell) = 1 - \mathcal{M}(Q - 1). \quad (4.26)$$

Combining this with (4.22) proves the claim. $\square$

Proposition 4.8. *For every fixed finite-field size Q ,*

$$\limsup_{n \rightarrow \infty} \delta_Q(n) = 1.$$

The upper limit is attained along the factorial extension degrees $n = M!$.

Proof. Let $M \rightarrow \infty$ through positive integers. If $r \leq M + 1$ is prime and $r \nmid Q$, Fermat's little theorem gives

$$\text{ord}_r(Q) \mid r - 1 \mid M!,$$

so $r \mid Q^{M!} - 1$. Lemma 4.6 therefore gives

$$\mathcal{M}(Q^{M!} - 1) \leq \prod_{\substack{r \leq M+1 \\ r \text{ prime, } r \nmid Q}} \left(1 - \frac{1}{r} + \frac{1}{r^2}\right). \quad (4.27)$$

Here $1 - 1/r + 1/r^2 = \mathcal{M}(r)$, and higher exponents of the prime divisors only decrease the product further.

To show that the right-hand side tends to zero, factor each term as

$$1 - \frac{1}{r} + \frac{1}{r^2} = \left(1 - \frac{1}{r}\right) \left(1 + \frac{1}{r(r-1)}\right). \quad (4.28)$$

All finite products of the second factors are uniformly bounded, since

$$\prod_r \left(1 + \frac{1}{r(r-1)}\right) \leq \exp \left(\sum_{j=2}^{\infty} \frac{1}{j(j-1)} \right) = e. \quad (4.29)$$

The product of the first factors tends to zero: for real $x \geq 2$, expanding the finite Euler product gives

$$\prod_{\substack{r \leq x \\ r \text{ prime}}} \left(1 - \frac{1}{r}\right)^{-1} \geq \sum_{j \leq x} \frac{1}{j} \rightarrow \infty. \quad (4.30)$$

Every prime divisor of a positive integer $j \leq x$ is at most x , so $1/j$ appears in the expansion on the left. Omitting the finitely many primes dividing the fixed integer Q changes only a constant factor. By (4.28)–(4.30), the right-hand side of (4.27) tends to zero, hence $\mathcal{M}(Q^{M!} - 1) \rightarrow 0$. Finally, (4.21) gives

$$\lim_{M \rightarrow \infty} \delta_Q(M!) = 1. \quad (4.31)$$

Since $\delta_Q(n) \leq 1$, this proves the stated upper limit. $\square$

Proof of Theorem 4.2. The degree, indeterminacy locus, and birationality follow from Lemmas 4.3 and 4.4; the finite-field count is Proposition 4.5. Propositions 4.7 and 4.8 prove (4.8). Since $\mathcal{M}(Q - 1) > 0$, the upper and lower limits are distinct. When $Q = 2$, we have $\mathcal{M}(Q - 1) = \mathcal{M}(1) = 1$, so the lower limit is 0. $\square$

Corollary 4.9. *Conjecture 18.10(b) of Benedetto et al. is false.*

Proof. Fix $\mathbb{P}^2/\mathbb{F}_2$ and the map (4.3). It is a dominant rational self-map, and Theorem 4.2 gives

$$\liminf_{n \rightarrow \infty} \frac{\#\mathbb{P}^2(\mathbb{F}_{2^n})_f}{\#\mathbb{P}^2(\mathbb{F}_{2^n})} = 0.$$

This contradicts the limit 1 required by the conjecture. $\square$

4.4 Geometric density over the algebraic closure

A set is Zariski dense if it is contained in no proper closed algebraic subset. In this example, density follows from the infinitely many initial values with everywhere-defined forward orbits for each fixed u . This holds alongside the oscillation of point-count proportions over finite-field extensions.

Proposition 4.10. *For every finite field $\mathbb{F}_Q$, the set $\mathbb{P}^2(\overline{\mathbb{F}}_Q)_f$ is Zariski dense in $\mathbb{P}^2$.*

Proof. Let $k = \overline{\mathbb{F}}_Q$. Every $u \in k^\times$ belongs to some finite field, so $\langle u \rangle$ is finite. Consider the following subset of the torus:

$$\mathcal{T} = \{(u, v) \in (k^\times)^2 : v \notin \langle u \rangle\}.$$

For $(u, v) \in \mathcal{T}$, all $u^j v - 1$ are nonzero and $v \neq 1$. Lemma 4.4 therefore gives $\Phi(\mathcal{T}) \subseteq \mathbb{P}^2(k)_f$.

A Laurent polynomial is a finite sum in which variables may have negative integer exponents. Suppose such a polynomial $H(u, v)$ vanishes on $\mathcal{T}$. For any fixed $u \in k^\times$, the Laurent polynomial $H(u, v)$ in v vanishes on $k^\times \setminus \langle u \rangle$. This set is obtained by removing finitely many values from the nonzero elements of an infinite field, so it is infinite. The Laurent polynomial is therefore zero. Expanding H in powers of v , each coefficient, viewed as a Laurent polynomial in u , vanishes on all of $k^\times$ and hence is zero. Thus $H = 0$, and $\mathcal{T}$ is Zariski dense in the torus. The coordinate isomorphism (4.9) sends this dense set into a dense open subset of $\mathbb{P}^2$, so $\Phi(\mathcal{T})$ is dense in the projective plane. $\square$

The geometric density asserted in Conjecture 18.10(a) of the survey therefore holds for this example. The counterexample in this section concerns the finite-field point-density assertion. The plane map constructed here is also distinct from the pentagram map in [9, Conjecture 8.3].

5 No universal seed for geometrically nilpotent subvarieties

For a morphism defined over a finite field, every point with values in the algebraic closure is defined over some finite extension, so its orbit is eventually periodic. We call these points geometric points in this section. If every geometric point of a subvariety eventually reaches the same fixed point, the time required may still depend on the point and have no uniform upper bound on the subvariety. In work announced in 2015 and published in 2018, Borisov constructed such geometrically nilpotent subvarieties that are not nilpotent. He asked whether one geometrically nilpotent subvariety could cover all others by taking forward iterates and inverse images [10, Definitions 2–3, Question 13]. This section gives a negative answer in every prime characteristic.

5.1 Definitions and main result

Let F be a finite field, $K = \overline{F}$, X a variety over F , and $T : X \rightarrow X$ a morphism over F . Subvarieties in this section may be reducible, and are assumed to be locally closed and reduced. We extend T to X_K and continue to denote it by T . Here X_K denotes the variety obtained by extending the field of definition from F to K , and $X(K)$ is its set of K -valued points; T^n denotes the n -fold iterate, with T^0 the identity.

Definition 5.1. A subvariety $Y \subseteq X_K$ is *geometrically nilpotent* with respect to T if there is a fixed point $P \in X(K)$ of T such that

$$\forall y \in Y(K) \quad \exists n_y \geq 1 : \quad T^{n_y}(y) = P.$$

If there is an integer $N \geq 1$ such that $T^N(Y) = \{P\}$, then Y is *nilpotent*. Neither definition requires Y to be T -invariant.

The definitions differ in the order of their quantifiers: geometric nilpotence allows each point its own absorption time n_y , whereas nilpotence requires one time N that works for all points. Since P is fixed, an orbit stays there after reaching P . For example, finitely many points that eventually reach P always have a uniform absorption time: take the maximum of their absorption times.

Borisov's question asks whether, for every T , there is a geometrically nilpotent subvariety Y , possibly empty or reducible, such that every geometrically nilpotent subvariety W satisfies

$$W \subseteq (T^\ell)^{-1}(T^m(Y)) \tag{5.1}$$

for some ℓ, m depending on W . A subvariety Y with this property is called a *universal seed*. For a given T , the quantifiers have the order

$$\exists Y \quad \forall W \quad \exists \ell, m \geq 0,$$

where Y, W are geometrically nilpotent subvarieties. Condition (5.1) is equivalent to $T^\ell(W) \subseteq T^m(Y)$: the same Y must accommodate every W , while the two iteration counts may vary with W . The proof below allows $\ell, m \geq 0$ and enlarges $T^m(Y)$ on the right-hand side to its Zariski closure.

Fix a prime p , set $K = \overline{\mathbb{F}_p}$, and define

$$\begin{aligned} D(x, y, z) &= (y - z)y(x - z), \\ T(x, y, z) &= (D(x, y, z)xz, D(x, y, z)xy, D(x, y, z)z^2). \end{aligned} \tag{5.2}$$

This is the reduction in characteristic p of a single homogeneous polynomial map of degree five with integer coefficients. For $s \geq 1$, write $q_s = p^s$ and set

$$Y_s = \{y^{q_s} = xz^{q_s-1}\} \subseteq \mathbb{A}_K^3. \tag{5.3}$$

Theorem 5.2. *The origin O is the unique fixed point of (5.2) over K . Each Y_s is a geometrically irreducible surface defined over $\mathbb{F}_p$ that is geometrically nilpotent and not nilpotent with respect to T . For every geometrically nilpotent subvariety $Y \subseteq \mathbb{A}_K^3$, all but finitely many s satisfy*

$$T^\ell(Y_s) \not\subseteq \overline{T^m(Y)} \quad \text{for all } \ell, m \geq 0. \tag{5.4}$$

Consequently, no subvariety Y as in (5.1) exists.

Images and inclusions in the theorem are interpreted on K -points, and a bar denotes Zariski closure in the relevant ambient variety. Allowing Y to be defined over K gives a larger class of candidates and therefore includes seeds defined over the original finite field. In the proof, the projection of each candidate Y has only finitely many curve or point components, while the projections of the surfaces for distinct s never coincide after any forward or backward iterates. They are distinguished by their function field extensions over the fixed ratio coordinate $u = x/z$.

5.2 The original map, the boundary, and all surviving orbits

Write $\mathbb{G}_m$ for the multiplicative group, so that $\mathbb{G}_m^2(K) = (K^\times)^2$. Set

$$U = \{xyz \neq 0\} \subseteq \mathbb{A}_K^3, \quad \pi : U \longrightarrow \mathbb{G}_m^2, \quad \pi(x, y, z) = (u, v) = (x/z, y/z).$$

In these coordinates,

$$D = v(v-1)(u-1)z^3.$$

If both the current point and its image lie in U , then (5.2) gives

$$(u', v') = (u, uv), \quad z' = v(v-1)(u-1)z^5. \quad (5.5)$$

Denote the torus automorphism by

$$S(u, v) = (u, uv). \quad (5.6)$$

Its integer powers are

$$S^j(u, v) = (u, u^j v), \quad j \in \mathbb{Z}. \quad (5.7)$$

The map S fixes the first coordinate u and multiplies the second coordinate by u at each step. The evolution of the ratio coordinates, together with the multiplier of the third coordinate in (5.5), determines whether the original orbit reaches the origin. In particular, if $u = 1$ or $v = 1$ at a point of U , then $D = 0$, and all three coordinates vanish at the next step.

Lemma 5.3. *Every point of $\mathbb{A}_K^3 \setminus U$ reaches O in at most two steps. In particular, for every $a \in \mathbb{A}^3(K)$ and $m \geq 0$,*

$$T^m(a) \in U \implies a, T(a), \dots, T^m(a) \in U. \quad (5.8)$$

Proof. If $y = 0$, then $D = 0$, and the next point is O . If $x = 0$, the first two coordinates of the image vanish, and the following point is O . If $z = 0$, direct substitution gives

$$T(x, y, 0) = (0, x^2 y^3, 0),$$

whose image is O . These cases also show that a point outside U can never enter U , proving (5.8). $\square$

Lemma 5.4. *The unique fixed point of T is O .*

Proof. Let (x, y, z) be a fixed point. If $z = 0$, the first coordinate gives $x = 0$, and the second then gives $y = 0$. If $z \neq 0$, the equation $Dz^2 = z$ implies $D \neq 0$, hence $y \neq 0$. The fixed-point equations $Dxy = y$ and $Dz^2 = z$ give $Dx = 1$ and $Dz = 1$, respectively. Thus $x = z$, contradicting $D \neq 0$. $\square$

Lemma 5.3 allows (5.5) to be iterated along every original orbit whose final point lies in U . More precisely,

$$\pi(T^m(a)) = S^m(\pi(a)) \quad \text{whenever } T^m(a) \in U. \quad (5.9)$$

The original map itself guarantees the required condition at all earlier steps.

5.3 Pointwise absorption without a uniform absorption time

The polynomial in (5.3), regarded as an element of $K[y, z][x]$, is primitive and linear in x : its coefficients z^{q_s-1} and y^{q_s} are coprime. It is irreducible in $K(y, z)[x]$, hence irreducible in $K[x, y, z]$ by Gauss's lemma. Thus Y_s is a geometrically irreducible surface. On the open set $z \neq 0$, it has the parametrization

$$(x, y, z) = (\tau^{q_s} z, \tau z, z). \quad (5.10)$$

Here $\tau \in K$ and $z \in K^\times$.

Proposition 5.5. *For every $s \geq 1$, all K -points of Y_s eventually reach O , but no single iterate sends the whole of Y_s to O .*

Proof. If $z = 0$, the surface equation forces $y = 0$, and the next point is O . If $z \neq 0$, use (5.10). When $\tau = 0$, we have $y = 0$; when $\tau = 1$, we have $x = z$. Both cases also reach O in one step.

Now suppose that $\tau \neq 0, 1$. Since $\tau \in K^\times$, it has finite multiplicative order M , with $\gcd(M, p) = 1$. Here the multiplicative order is the smallest positive integer M such that $\tau^M = 1$; it divides the order of the multiplicative group of some finite field and is therefore coprime to p . As long as the orbit has not left U , (5.7) gives

$$u_n = \tau^{q_s}, \quad v_n = \tau^{1+nq_s}. \quad (5.11)$$

Since q_s is invertible in $\mathbb{Z}/M\mathbb{Z}$, there is an $n \geq 0$ such that $1 + nq_s \equiv 0 \pmod{M}$. If the orbit has already left U , Lemma 5.3 ensures that it eventually reaches O . Otherwise, $v_n = 1$ at this time, and the factor $v_n - 1$ in (5.5) sends the point in the original coordinates to O at the next step.

On the other hand, fix any $N \geq 0$. Since K is infinite, we can choose $\tau \in K^\times$ avoiding the roots of the following finitely many equations:

$$\tau^{q_s} = 1, \quad \tau^{1+jq_s} = 1 \quad (0 \leq j \leq N). \quad (5.12)$$

Each of the polynomials involved is nonzero. Taking $z = 1$, equations (5.5) and (5.11) show that this point remains in U through its N th iterate, so $T^N(Y_s) \neq \{O\}$. This holds for every N . $\square$

For example, take $p = 2$ and $s = 1$, and let $\omega \in \mathbb{F}_4^\times$ have order 3. The surface point $(\omega^2, \omega, 1)$ has initial ratios $(u_0, v_0) = (\omega^2, \omega)$, and its first step gives $v_1 = \omega^{1+2} = 1$. Direct substitution into the original map yields

$$T(\omega^2, \omega, 1) = (1, \omega, \omega), \quad T^2(\omega^2, \omega, 1) = O.$$

The generic point of the surface has two algebraically independent coordinate parameters τ, z , with values in the function field $K(\tau, z)$, the field of rational functions in these two variables. In this field, $\tau^{q_s} - 1$ and every $\tau^{1+nq_s} - 1$ are nonzero, so the orbit of the generic point stays in U . The definition of geometric nilpotence ranges over $Y_s(K)$; the generic point has values in a larger field. The two assertions of the proposition reflect, respectively, finite multiplicative order after specialization to a finite field and the failure of every fixed iterate to contract the entire surface identically.

Define the torus curves

$$C_s = \{u = v^{q_s}\} \subseteq \mathbb{G}_m^2. \quad (5.13)$$

The parametrization $\tau \mapsto (\tau^{q_s}, \tau)$ is an isomorphism from $\mathbb{G}_m$ to C_s . Thus each C_s is a rational curve, meaning that its function field is a rational function field in one variable. The comparisons below retain the projection $(u, v) \mapsto u$. The induced function field extension has degree p^s ; this degree and its pure inseparability are discussed in the last subsection.

Lemma 5.6. *For every $s \geq 1$ and $n \geq 0$,*

$$\overline{\pi(T^n(Y_s) \cap U)} = S^n(C_s), \quad (5.14)$$

where the closure is taken in $\mathbb{G}_m^2$.

Proof. By (5.8), each point of $T^n(Y_s) \cap U$ is the image of an orbit that has stayed in U . Thus (5.9) gives the inclusion of the set on the left in $S^n(C_s)$.

Conversely, choose τ outside the finite set excluded by (5.12) with $N = n$, and take any $z \in K^\times$. The corresponding initial point stays in U for the first n steps, and its projection is

$$(\tau^{q_s}, \tau^{1+nq_s}) = S^n(\tau^{q_s}, \tau).$$

These parameters form a dense open subset of $\mathbb{G}_m$. Since S^n is an automorphism, their image is dense in $S^n(C_s)$, which proves the claim. $\square$

5.4 Projection dimension of an arbitrary geometrically nilpotent subvariety

First consider the set of torus points

$$E = \{(u, v) \in (K^\times)^2 : u \neq 1, v \notin \langle u \rangle\}, \quad (5.15)$$

where $\langle u \rangle$ is the finite cyclic subgroup of $K^\times$ generated by u .

Lemma 5.7. *The set E is Zariski dense in $\mathbb{G}_m^2$. For every $(u, v) \in E$ and every $z \in K^\times$, all iterates of (uz, vz, z) lie in U , so this point never reaches O .*

Proof. For such an initial point, $u_n = u$ and $v_n = u^n v$. Since $v \notin \langle u \rangle$, we have $v_n \neq 1$ for every $n \geq 0$. The multiplier of the third coordinate in (5.5) is therefore nonzero at every step, and the assertion follows by induction.

Suppose that a Laurent polynomial vanishes on E . After multiplying by a monomial, we may regard it as a polynomial in $K[u, v]$. Fix any $u \in K^\times \setminus \{1\}$. The permitted values of v exclude only the finite set $\langle u \rangle$, so the polynomial in v is identically zero. Each of its coefficients as a polynomial in v therefore vanishes at infinitely many u , and is itself identically zero. This proves the density of E . $\square$

The following lemma ensures that the image of a dominant morphism contains a nonempty open set, allowing an escaping point in the projection to be lifted to a point of the subvariety.

Lemma 5.8. *Let K be an algebraically closed field, and let $f : V \rightarrow W$ be a dominant morphism of irreducible affine K -varieties, meaning that its image is dense in W . Then $f(V(K))$ contains a nonempty Zariski open subset of $W(K)$.*

Proof. This is a case of Chevalley's theorem and also admits a direct proof with coordinate rings. Write $A = K[W] \subseteq B = K[V]$ and $L = \text{Frac}(A)$. By Zariski's lemma, a maximal ideal of the nonzero finitely generated L -algebra $B \otimes_A L$ gives a finite field extension L'/L and an A -algebra homomorphism $B \rightarrow L'$.

Choose finitely many generators of B as an A -algebra. Their images in L' are algebraic over L . Clearing denominators in monic equations for these elements gives $0 \neq g \in A$ such that all the images are integral over A_g . Let C be the A_g -algebra generated by these images. Then $A_g \subseteq C$ is a finite integral extension, and there is a surjection $B_g \rightarrow C$. Integral extensions satisfy lying-over: every maximal ideal of A_g has a maximal ideal of C above it. The corresponding residue field is finite over K and hence equals K because K is algebraically closed. Pulling back along $B_g \rightarrow C$ gives a point of $V(K)$. Thus all K -points of the nonempty principal open subset $D(g) = \{w \in W : g(w) \neq 0\}$ of W lie in the actual image. $\square$

Lemma 5.9. *Let $W \subseteq \mathbb{A}_K^3$ be an irreducible geometrically nilpotent subvariety with $W \cap U \neq \emptyset$. Then*

$$C_W = \overline{\pi(W \cap U)}$$

is an irreducible closed subvariety of dimension at most one.

Proof. The image closure is irreducible. If $C_W = \mathbb{G}_m^2$, then

$$\pi|_{W \cap U} : W \cap U \longrightarrow \mathbb{G}_m^2$$

is dominant. Choose a nonempty affine open subvariety V of $W \cap U$. The restriction to V remains dominant: if its image were contained in a closed set, the inverse image of that closed set would contain the dense subset V and hence all of $W \cap U$. By Lemma 5.8, the actual image on K -points contains a nonempty open set. Lemma 5.7 gives a point $(u, v) \in E$ in that open set. Choose a lift (uz, vz, z) in $W \cap U$. This point never reaches O , while Lemma 5.4 shows that O is the unique fixed point, contradicting geometric nilpotence of W .

Thus C_W is a proper irreducible closed subvariety of the irreducible two-dimensional torus, so it has dimension at most one. $\square$

5.5 Curve classes and arbitrary forward and backward iterates

By (5.7), for every $j \in \mathbb{Z}$,

$$S^j(C_s) = \{v^{q_s} = u^{1+jq_s}\}. \quad (5.16)$$

Negative exponents are interpreted in the Laurent coordinate ring of the torus.

Lemma 5.10. *If $s, r \geq 1$ and $j, h \in \mathbb{Z}$ satisfy*

$$S^j(C_s) = S^h(C_r),$$

then $s = r$ and $j = h$.

Proof. Substitute the parametrization of the curve on the left,

$$(u, v) = (\tau^{q_s}, \tau^{1+jq_s}), \quad \tau \in K^\times,$$

into the equation of the curve on the right. This gives

$$\tau^{q_r(1+jq_s)} = \tau^{q_s(1+hq_r)} \quad \text{for every } \tau \in K^\times.$$

Distinct integer powers cannot agree identically over an infinite field, so

$$q_r(1+jq_s) = q_s(1+hq_r), \quad j-h = \frac{1}{q_r} - \frac{1}{q_s}.$$

The left-hand side is an integer and the right-hand side has absolute value less than one. Hence $j = h$, and then $q_s = q_r$. This comparison of exponents takes place in the integers and follows from the identity of Laurent power functions. $\square$

Proof of Theorem 5.2. The preceding results establish uniqueness of the fixed point, geometric irreducibility of Y_s , pointwise absorption, and the absence of a uniform absorption time. Fix an arbitrary geometrically nilpotent subvariety Y .

If Y is empty, (5.4) holds for every s . Otherwise, write Y as the union of its finitely many irreducible components, $Y = W_1 \cup \dots \cup W_d$. Each component is geometrically nilpotent. Discard the components disjoint from U , and for the remaining components set

$$B_i = \overline{\pi(W_i \cap U)}.$$

Lemma 5.9 gives $\dim B_i \leq 1$. For $m \geq 0$, put

$$B^{(m)} = \bigcup_i S^m(B_i) \subseteq \mathbb{G}_m^2.$$

This is a finite union of closed subvarieties.

If an actual image point $T^m(y)$ lies in U , Lemma 5.3 ensures that y and all its earlier iterates lie in U . Hence

$$T^m(Y) \cap U \subseteq \pi^{-1}(B^{(m)}). \quad (5.17)$$

The right-hand side is closed in U . Since U is open in $\mathbb{A}_K^3$,

$$\overline{T^m(Y)} \cap U = \overline{T^m(Y) \cap U}^U \subseteq \pi^{-1}(B^{(m)}).$$

This controls both the actual image and its closure.

Suppose that, for some s , there exist $\ell, m \geq 0$ such that $T^\ell(Y_s) \subseteq \overline{T^m(Y)}$. Intersecting with U , projecting, taking closure, and applying Lemma 5.6 gives

$$S^\ell(C_s) \subseteq \bigcup_i S^m(B_i). \quad (5.18)$$

The left-hand side is an irreducible curve, and every closed subvariety on the right has dimension at most one. Thus some one-dimensional B_i satisfies

$$S^\ell(C_s) = S^m(B_i), \quad B_i = S^{\ell-m}(C_s).$$

By Lemma 5.10, a fixed B_i can correspond to at most one s , regardless of the choice of ℓ, m . There are only finitely many B_i , so only finitely many Y_s can be covered by any such forward and backward iterates. This proves (5.4). $\square$

5.6 Purely inseparable degree over the fixed base

The curve classes above can be interpreted as curves (C_s, u) equipped with a projection. Here $K(C_s)$ is the field of rational functions on the curve, and the parameter τ is transcendental in the function field. Fixing the projection $(u, v) \mapsto u$ onto the first coordinate axis gives the subfield and extension

$$K(u) = K(\tau^{p^s}) \subseteq K(C_s) = K(\tau). \quad (5.19)$$

An algebraic extension L/E in characteristic p is *purely inseparable* if, for every $\alpha \in L$, there is an $e \geq 0$ such that $\alpha^{p^e} \in E$. In (5.19), we have $\tau^{p^s} = u$, and K is algebraically closed, so the extension is purely inseparable. Its degree is p^s : after clearing denominators and grouping powers of τ by their exponents modulo p^s , we see that $1, \tau, \dots, \tau^{p^s-1}$ are linearly independent over $K(\tau^{p^s})$, while $\tau^{p^s} = u$ ensures that they span the extension.

Since the automorphism S fixes u , the purely inseparable degree of (5.19) is preserved by every S^j . More explicitly, $S^j(C_s)$ has the parametrization $(u, v) = (\tau^{p^s}, \tau^{1+jp^s})$, and $\tau = v/u^j$. Its function field is therefore still $K(\tau)$, with the same base subfield $K(u)$. All the C_s are isomorphic to $\mathbb{G}_m$ as abstract curves; the degree p^s records their projections onto the same u -axis.

For example, in characteristic 2, the curves $u = v^2$ and $u = v^4$ both have function fields isomorphic to $K(\tau)$, while the extensions over the fixed coordinate u have degrees 2 and 4, respectively. On points over any finite extension field, the map $\tau \mapsto \tau^{p^s}$ is a bijection, since Frobenius is an automorphism of a finite field. This pointwise invertibility gives the absorption times in (5.11); the infinitely many function field degrees, through (5.18), restrict the surfaces that a single seed of finite type can cover.

The following criterion extends this relation to surfaces with monomial parametrizations. An integer exponent pair (a, b) is *primitive* if $\gcd(|a|, |b|) = 1$. This condition ensures that the parameter can be recovered as a product of integer powers of u, v .

Proposition 5.11. *Let $a, b \in \mathbb{Z}$ satisfy $\gcd(|a|, |b|) = 1$, and define*

$$V_{a,b} = \overline{\{(\tau^a z, \tau^b z, z) : \tau, z \in K^\times\}} \subseteq \mathbb{A}_K^3.$$

If $a = 0$, then $V_{a,b} = \{x = z\}$ and reaches O in one step. If $a \neq 0$, then $V_{a,b}$ is geometrically nilpotent if and only if $|a| = p^s$ for some $s \geq 0$.

Proof. Choose integers i, j such that $ia + jb = 1$. The image of the parametrization in ratio coordinates is exactly the closed torus curve $u^b v^{-a} = 1$: on this curve, set $\tau = u^i v^j$; then $\tau^a = u$ and $\tau^b = v$. Thus the cone inside U is also given exactly by the stated parametrization. Boundary points are covered by Lemma 5.3.

If $a = 0$, then $b = \pm 1$, and the conclusion follows from the factor $x - z$. If $|a| = p^s$, take any parameter $\tau \in K^\times$ and let M be its finite multiplicative order. Then a is invertible in $\mathbb{Z}/M\mathbb{Z}$. As long as absorption has not occurred, the ratio coordinates satisfy

$$u_n = \tau^a, \quad v_n = \tau^{b+na}.$$

Choose $n \geq 0$ such that $b + na \equiv 0 \pmod{M}$. Then $v_n = 1$, and the next point is O .

Conversely, suppose that a prime $r \neq p$ divides a , and let $d = v_r(|a|) \geq 1$, so that $r^d \mid a$ and $r^{d+1} \nmid a$. Choose $h > d$ and take $\tau \in K^\times$ of order r^h . Primitivity gives $r \nmid b$, so

$$\text{ord}(\tau^a) = r^{h-d} > 1, \quad \text{ord}(\tau^b) = r^h.$$

Hence $\tau^a \neq 1$ and $\tau^b \notin \langle \tau^a \rangle$. By Lemma 5.7, the geometric point obtained by taking $z = 1$ never reaches O , contradicting geometric nilpotence. $\square$

When $a \neq 0$, the projection to u corresponds to $K(\tau^a) \subseteq K(\tau)$. Write $|a| = p^s a_0$, where $a_0 > 0$ and $p \nmid a_0$. The purely inseparable degree is p^s , and the separable degree is a_0 . The separable degree is the number of distinct embeddings of this finite extension into an algebraic closure that fix the base field; the total degree is the product of the separable and purely inseparable degrees. Thus Proposition 5.11 states that these primitive monomial surfaces are geometrically nilpotent exactly when the corresponding function field extension is purely inseparable.

For example, in characteristic 2, the surface $V_{2,1} = Y_1$ is geometrically nilpotent, while $V_{3,1}$ contains points that never reach O . Take $\tau \in \overline{\mathbb{F}_2}^\times$ of order 9. Then $u = \tau^3$ has order 3, and $v = \tau$ has order 9, so $v \notin \langle u \rangle$. Lemma 5.7 therefore applies to the point $(\tau^3, \tau, 1)$ of the original three-dimensional system.

6 Formal verification and code

The Lean 4 source, dependency lock files, and reproduction commands are available in the repository

[https://github.com/veridiscoversLab/
arithmetic-dynamics-realization-extension-obstructions](https://github.com/veridiscoversLab/arithmetic-dynamics-realization-extension-obstructions)

This section refers to commit [e127cea](#), verified with Lean 4.29.0 and the pinned Mathlib version. The [formalization scope document](#) lists the declarations and source files.

End-to-end formalized results. The following results are formalized from their stated algebraic data and definitions, with all required intermediate results proved in Lean.

- (1) The inverse realization statement of Theorem 2.2, through [theorem22](#). For every prime p and every p -normal set, it constructs a constant-coefficient linear recurrence over the fixed field $\mathbb{F}_p(t)$ and proves equality of its zero set with the prescribed set at every natural time. The statement includes rational parameters, mixed signs, and finite modifications.
- (2) The exact count (4.6) and density extrema (4.8) in Theorem 4.2, including failure of convergence to one over $\mathbb{F}_2$. The [corresponding declarations](#) use partial maps on finite-field projective points, define each successive iterate of the original map, and count the points whose orbit is defined at every step. Proposition 4.10 is also verified in its formulation through vanishing polynomials.
- (3) The core dynamical nonexistence statement for a universal geometrically nilpotent seed in Theorem 5.2, through [theorem_5_2'](#) and [theorem_5_2_locallyClosed](#). This formulation uses algebraic-closure-valued points and polynomial zero sets. It retains the original map T , every geometrically nilpotent locally closed seed, and all iteration quantifiers in (5.4). The monomial absorption criterion in Proposition 5.11 is formalized as well.

Scope of projective extension. The declaration [theorem_3_1_projective](#) takes a homogeneous ideal I with a nonempty geometric projective zero locus and a tuple of forms of degree $d \geq 2$ that preserves I and has no common base point on this locus. It proves a closed immersion of schemes, twisting-sheaf pullback isomorphisms, an ambient extension of the same degree, and commuting diagrams for all iterates. The passage from arbitrary original polarized data $(X, \mathcal{L}, \phi, \mu)$ to these coordinate inputs remains to be formalized. It includes constructing an embedding from an ample power, identifying the original tensor powers, proving surjectivity of section restriction ([hres](#) in the code), and identifying the original scheme and morphism. The general scheme statement of Theorem 3.1 therefore has a remaining gap in its end-to-end formalization. These steps are absent from the proof dependencies of the three results listed above.

Several ancillary statements remain outside the formalized scope: the forward classification quoted from Derksen in Corollary 2.8; the degree-one boundary in Proposition 3.7; the absence of a nonconstant common coordinate factor, the minimal representation degree, and the identification of coordinate certificates with scheme-theoretic rational maps in Section 4; and the dimension and function-field degree statements in Section 5. The repository build and axiom audit passed at the stated version. The axiom audit reports only [propext](#), [Classical.choice](#), and [Quot.sound](#), expressing propositional extensionality, classical choice, and soundness of quotient types. The proof source contains no [sorry](#) or custom mathematical axioms.

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